MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
49PQ POWER DISTRIBUTION: PROCESS PLANT SUBSTATIONS
FOUNDATION FOR SS - 03 & 04

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

This document is to Prepare for design of Foundation for **Prefab Substation (49PQ) SS-03 & 04**. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel. For member strength calculation force values are directly taken from Load Data Provided.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Foundation and Civil Design Input attached in ATTACHMENT-1. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5 CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar: ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

5.3 Foundations for Prefab Substation.

Foundation shall be Isolated or Combined Foundation to support the structural framing of prefabricated substation.

5.4 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Raft & Column	50

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5.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450 kN/m^2 at 1.5m depth below EGL (ie. RL 912.25m) refer Table 4 - BH-19 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

6. DESIGN CALCULATION FOR FOUNDATION

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MD-216-49PQ-EG-CV-CAL-1625	Mansourah Massarah Gold Project	
900707_GAU01_WD01_CED01_ 45006_OU100684606_R2	DESIGN CALCULATION POWER DISTRIBUTION: PROCESS PLANT SUBSTATIONS FOUNDATION FOR SS – 03 & 04	Date: 25.10.2023
O19045_950_C_K_24310_001_R2	AS BUILT	Rev: D01

DESIGN OF FOUNDATION MKD. F1 of SS#3

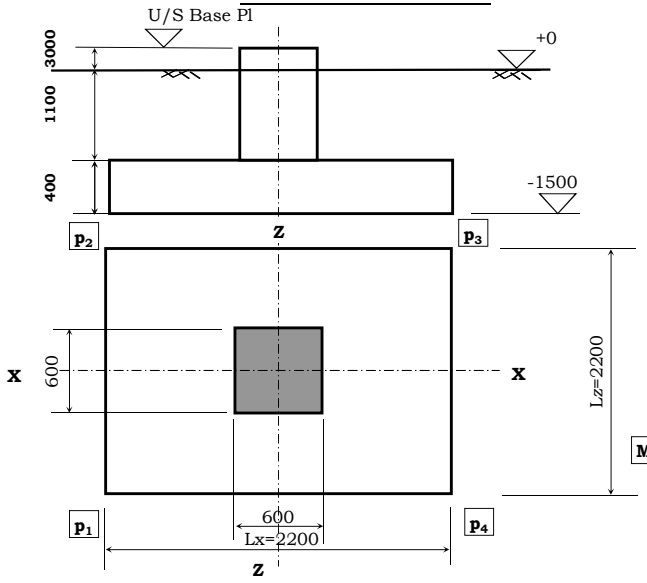
	Along-X	Along-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	2.20m	2.20m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	3.00 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

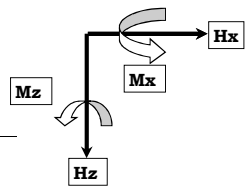
SYMBOLS AND NOTATIONS

Dead Load	DL
Live Load	LL
Wind Load (X)	WLx
Wind Load (Z)	WLz
Earthquake Load (X)	ELx
Earthquake Load (Z)	ELz
Live Load (Roof)	LLr

Calculation of Weight of Foundation :

Weight of raft & Col.=	(2.2*2.2*0.4*24)+(0.6*0.6*4.1*24) =	81.89KN
Weight of soil =	((2.2*2.2)-(0.6*0.6))*1.1*18 =	88.70KN
	Total =	170.59KN





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 344mm = 0.344m

Effective depth of slab along Z direction = 338mm = 0.338m

Clear projection along Z direction = 0.8m

Clear projection along X direction = 0.8m

H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F1 of SS#3

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)
1	Dead Load (DL)	35.18	0.66	0.21	10.66	0.18
2	Live Load (LL)	49.48	3.60	-3.63	24.40	0.49
3	Wind Load (WLx)	-9.91	2.64	-1.67	-0.67	0.16
4	Wind Load (WLz)	-6.67	-0.44	-0.30	-0.01	-0.01
5	Earthquake Load (ELx)	-0.07	-2.69	0.19	0.01	0.00
6	Earthquake Load (ELz)	-1.15	0.66	-0.46	0.65	0.05
7	Live Load (Roof)(LLr)	9.61	-0.79	0.75	0.86	-0.02

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	DL+LL	84.66	4.26	-3.42	35.06	0.67	255.25	4.26	15.76	35.06	158.42
202	DL+0.75LL+0.75LLr	79.50	2.77	-1.96	29.60	0.53	250.09	2.77	10.53	29.60	133.74
203	DL+WLx	25.28	3.30	-1.46	9.99	0.34	195.87	3.30	13.37	9.99	45.28
204	DL+WLz	28.51	0.22	-0.09	10.65	0.17	199.10	0.22	0.92	10.65	48.09
205	DL+0.7ELx	35.13	-1.22	0.34	10.67	0.18	205.72	-1.22	-5.17	10.67	48.18
206	DL+0.7ELz	34.38	1.12	-0.12	11.12	0.22	204.97	1.12	4.94	11.12	50.25
207	DL+0.75WLx+0.75LL+0.75LLr	72.07	4.75	-3.21	29.10	0.65	242.66	4.75	18.17	29.10	131.59
208	DL+0.75WLz+0.75LL+0.75LLr	74.49	2.45	-2.18	29.60	0.52	245.08	2.45	8.83	29.60	133.70
209	DL+0.525ELx+0.75LL+0.75LLr	79.46	1.36	-1.86	29.61	0.53	250.05	1.36	4.26	29.61	133.77
210	DL+0.525ELz+0.75LL+0.75LLr	78.89	3.12	-2.20	29.95	0.56	249.48	3.12	11.85	29.95	135.32
211	0.6DL + WLx	11.20	3.03	-1.54	5.72	0.27	181.79	3.03	12.10	5.72	26.02
212	0.6DL + WLz	14.44	-0.04	-0.17	6.39	0.10	185.03	-0.04	-0.36	6.39	28.83
213	0.6DL + 0.7ELx	21.06	-1.49	0.26	6.40	0.11	191.65	-1.49	-6.45	6.40	28.92
214	0.6DL + 0.7ELx	20.31	0.86	-0.20	6.85	0.15	190.90	0.86	3.67	6.85	30.99

Ultimate Conditions

301	1.4 DL	49.25	0.93	0.29	14.92	0.25	219.84	0.93	4.45	14.92	67.41
302	1.4 DL + 1.7 LL +0.5LLr	138.17	6.66	-5.51	56.83	1.07	308.76	6.66	24.45	56.83	256.80
303	1.2 DL+1.6WLx+LL +0.5LLr	80.65	8.22	-5.68	36.54	0.95	251.24	8.22	31.31	36.54	165.39
304	1.2 DL+1.6WLz+LL +0.5LLr	85.82	3.30	-3.48	37.60	0.68	256.41	3.30	11.38	37.60	169.89
305	1.2 DL+ELx+LL	91.62	1.70	-3.19	37.20	0.70	262.21	1.70	4.46	37.20	168.10
306	1.2 DL+ELz+LL	90.55	5.06	-3.84	37.84	0.76	261.14	5.06	18.91	37.84	171.05
307	0.9DL+ 1.6WLx	15.82	4.81	-2.48	8.52	0.41	186.41	4.81	19.17	8.52	38.74
308	0.9DL+ 1.6WLz	20.99	-0.10	-0.29	9.58	0.14	191.58	-0.10	-0.76	9.58	43.25
309	0.9DL+ ELx	31.59	-2.10	0.37	9.60	0.16	202.18	-2.10	-9.07	9.60	43.38
310	0.9DL+ Elz	30.52	1.26	-0.27	10.25	0.22	201.11	1.26	5.38	10.25	46.33

Check for overturning and sliding:

Load Cases:		Load at B/O Fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	DL+LL	255.25	4.26	15.76	35.06	158.42	17.81	1.77	23.95	2.91
202	DL+0.75LL+0.75LLr	250.09	2.77	10.53	29.60	133.74	26.14	2.06	36.07	3.38
203	DL+WLx	195.87	3.30	13.37	9.99	45.28	16.11	4.76	23.77	7.85
204	DL+WLz	199.10	0.22	0.92	10.65	48.09	239.20	4.55	356.23	7.48
205	DL+0.7ELx	205.72	-1.22	-5.17	10.67	48.18	43.74	4.70	67.19	7.71
206	DL+0.7ELz	204.97	1.12	4.94	11.12	50.25	45.64	4.49	72.98	7.37
207	DL+0.75WLx+0.75LL+0.75LLr	242.66	4.75	18.17	29.10	131.59	14.69	2.03	20.43	3.34
208	DL+0.75WLz+0.75LL+0.75LLr	245.08	2.45	8.83	29.60	133.70	30.54	2.02	40.08	3.31
209	DL+0.525ELx+0.75LL+0.75LLr	250.05	1.36	4.26	29.61	133.77	64.56	2.06	73.57	3.38
210	DL+0.525ELz+0.75LL+0.75LLr	249.48	3.12	11.85	29.95	135.32	23.17	2.03	31.98	3.33
211	0.6DL + WLx	181.79	3.03	12.10	5.72	26.02	16.53	7.69	23.98	12.71
212	0.6DL + WLz	185.03	-0.04	-0.36	6.39	28.83	571.15	7.06	1814.51	11.59
213	0.6DL + 0.7ELx	191.65	-1.49	-6.45	6.40	28.92	32.71	7.29	51.48	11.97
214	0.6DL + 0.7ELx	190.90	0.86	3.67	6.85	30.99	57.25	6.78	88.89	11.14

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DESIGN OF FOUNDATION MKD. F1 of SS#3

Load Combination & Calculation of Base pressure

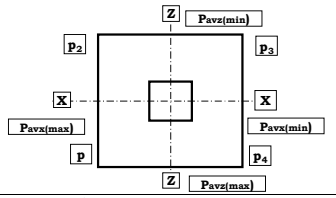
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m²

Net SBC of soil for DL, LL= 450 kN/m² For WL/SL case = 562.5 kN/m²

Gross SBC of soil for DL, LL= 477 kN/m² For WL/SL case = 589.5 kN/m²



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m²)													
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p1 (pmax)	p2	p4	p3 (pmin)	(pmax-γh)	Pavx(max)	Pavx(min)	Pavz(max)	Pavz(min)	Remarks
201	DL+LL	255.25	15.76	158.42	52.74	8.88	89.27	180.00	0.00	180.00	0.00	153.00	90.00	90.00	180.00	0.00	O.K
202	DL+0.75LL+0.75LLr	250.09	10.53	133.74	51.67	5.93	75.36	175.00	0.00	175.00	0.00	148.00	87.50	87.50	175.00	0.00	O.K
203	DL+WLx	195.87	13.37	45.28	40.47	7.54	25.51	73.52	22.49	58.45	7.42	46.52	48.01	32.94	65.99	14.96	O.K
204	DL+WLz	199.10	0.92	48.09	41.14	0.52	27.10	68.75	14.55	67.72	13.52	41.75	41.65	40.62	68.24	14.04	O.K
205	DL+0.7ELx	205.72	5.17	48.18	42.50	2.92	27.15	72.57	18.27	66.74	12.44	45.57	45.42	39.59	69.66	15.36	O.K
206	DL+0.7ELz	204.97	4.94	50.25	42.35	2.78	28.31	73.45	16.82	67.88	11.25	46.45	45.14	39.57	70.67	14.04	O.K
207	DL+0.75WLx+0.75LL+0.75LLr	242.66	18.17	131.59	50.14	10.24	74.15	150.00	0.00	150.00	0.00	123.00	75.00	75.00	150.00	0.00	O.K
208	DL+0.75WLz+0.75LL+0.75LLr	245.08	8.83	133.70	50.64	4.97	75.34	150.00	0.00	150.00	0.00	123.00	75.00	75.00	150.00	0.00	O.K
209	DL+0.525ELx+0.75LL+0.75LLr	250.05	4.26	133.77	51.66	2.40	75.38	155.00	0.00	155.00	0.00	128.00	77.50	77.50	155.00	0.00	O.K
210	DL+0.525ELz+0.75LL+0.75LLr	249.48	11.85	135.32	51.55	6.68	76.25	155.00	0.00	155.00	0.00	128.00	77.50	77.50	155.00	0.00	O.K
211	0.6DL + WLx	181.79	12.10	26.02	37.56	6.82	14.66	59.04	29.72	45.40	16.08	32.04	44.38	30.74	52.22	22.90	O.K
212	0.6DL + WLz	185.03	0.36	28.83	38.23	0.20	16.25	54.68	22.18	54.28	21.78	27.68	38.43	38.03	54.48	21.98	O.K
213	0.6DL + 0.7ELx	191.65	6.45	28.92	39.60	3.63	16.30	59.53	26.93	52.26	19.67	32.53	43.23	35.97	55.90	23.30	O.K
214	0.6DL + 0.7ELx	190.90	3.67	30.99	39.44	2.07	17.46	58.97	24.05	54.84	19.91	31.97	41.51	37.38	56.91	21.98	O.K
301	1.4 DL	219.84	4.45	67.41	45.42	2.51	37.98	85.91	9.95	80.90	4.93	58.91	47.93	42.92	83.41	7.44	O.K
302	1.4 DL + 1.7 LL +0.5LLr	308.76	24.45	256.80	63.79	13.78	144.70	250.00	0.00	260.00	0.00	223.00	125.00	130.00	255.00	0.00	O.K
303	1.2 DL+1.6WLx+LL +0.5LLr	251.24	31.31	165.39	51.91	17.64	93.19	190.00	0.00	200.00	0.00	163.00	95.00	100.00	195.00	0.00	O.K
304	1.2 DL+1.6WLz+LL +0.5LLr	256.41	11.38	169.89	52.98	6.41	95.73	190.00	0.00	180.00	0.00	163.00	95.00	90.00	185.00	0.00	O.K
305	1.2 DL+ELx+LL	262.21	4.46	168.10	54.18	2.52	94.72	200.00	0.00	180.00	0.00	173.00	100.00	90.00	190.00	0.00	O.K
306	1.2 DL+ELz+LL	261.14	18.91	171.05	53.95	10.66	96.39	200.00	0.00	200.00	0.00	173.00	100.00	100.00	200.00	0.00	O.K
307	0.9DL+ 1.6WLx	186.41	19.17	38.74	38.51	10.80	21.83	71.15	27.49	49.54	5.88	44.15	49.32	27.71	60.35	16.69	O.K
308	0.9DL+ 1.6WLz	191.58	0.76	43.25	39.58	0.43	24.37	64.38	15.64	63.52	14.78	37.38	40.01	39.15	63.95	15.21	O.K
309	0.9DL+ ELx	202.18	9.07	43.38	41.77	5.11	24.44	71.33	22.44	61.11	12.22	44.33	46.89	36.67	66.22	17.33	O.K
310	0.9DL+ Elz	201.11	5.38	46.33	41.55	3.03	26.11	70.69	18.47	64.63	12.41	43.69	44.58	38.52	67.66	15.44	O.K
	Max :	255	18	158													
	Min :	196	1	45													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 153.00 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 163.00 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 148.00 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 158.00 kN/m²

Sample calculation of values of base pressure:

Load case : DL+0.75LL+0.75LLr

Maximum pressure, p_1 (pmax) = 51.67+5.93+75.36= 132.96 kN/m²

p_2= 51.67+5.93-75.36= -17.76 kN/m²

p_3= 51.67-(5.93+75.36)= -29.62 kN/m²

p_4= 51.67-5.93+75.36= 121.10 kN/m²

Maxm pressure excluding weight of soil (p_{max}-γh)= 175.00-1.5*18= 148.00 kN/m²

Maximum average Pressure along X, $p_{avx(max)}$ = (175.00+0.00)/2= 57.60 kN/m²

Minimum average Pressure along X, $p_{avx(minx)}$ = (0.00+175.00)/2= 45.74 kN/m²

Maximum average Pressure along Y, $p_{avy(max)}$ = (175.00+175.00)/2 127.03 kN/m²

Minimum average Pressure along Y, $p_{avy(minx)}$ = (0.00+0.00)/2= -23.69 kN/m²

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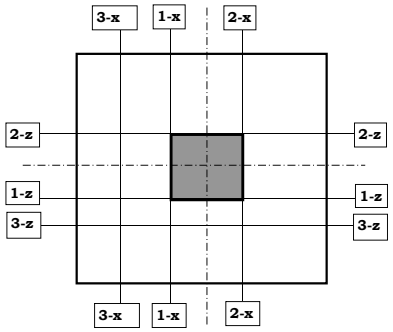
DESIGN OF FOUNDATION MKD. F1 of SS#3

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.
p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.
p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.
BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.
BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.
SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.6+2*0.5*0.338)+(0.6+2*0.5*0.338))*2
= 3.752 m
Punching Area= ((0.6+2*0.5*0.338)*(0.6+2*0.5*0.338))
= 0.88 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.1*18)
= 29.40 kN/m^2
Area of footing, A_f= 4.84 m^2



Load Case	Load Combination	p_{1x}	p_{2x}	p_{3x}	B.M$_{1x}$	B.M$_{2x}$	S.F$_{3x}$	S.F$_{(punching)}$
Ultimate Conditions								
301	1.4 DL	46.11	44.74	46.89	5.74	4.52	8.21	63.45
302	1.4 DL + 1.7 LL +0.5LLr	126.82	128.18	126.04	30.79	32.00	43.83	136.20
303	1.2 DL+1.6WLx+LL +0.5LLr	96.82	98.18	96.04	21.19	22.40	30.15	89.14
304	1.2 DL+1.6WLz+LL +0.5LLr	93.18	91.82	93.96	20.80	19.59	29.68	93.37
305	1.2 DL+ELx+LL	96.36	93.64	97.93	22.20	19.78	31.72	98.11
306	1.2 DL+ELz+LL	100.00	100.00	100.00	22.59	22.59	32.19	97.24
307	0.9DL+ 1.6WLx	41.46	35.57	44.84	5.54	0.30	8.06	36.09
308	0.9DL+ 1.6WLz	39.70	39.46	39.83	3.36	3.15	4.80	40.32
309	0.9DL+ ELx	43.17	40.39	44.77	5.20	2.72	7.49	49.00
310	0.9DL+ Elz	42.38	40.72	43.32	4.62	3.15	6.64	48.12

Sample calculation of BM & SF

Load case : 301

p_{1x} = 45.14-(45.14-39.57)*0.8/2.2 = 43.11 kN/m^2 Overhang = 0.8 m
p_{2x} = 39.57+(45.14-39.57)*0.8/2.2 = 41.60 kN/m^2 Effective depth = 0.344 m
p_{3x} = 45.14-(45.14-39.57)*(0.8-0.344)/2.2 = 43.99 kN/m^2
BM_{1x} = (46.11-29.4)*0.8*0.8/2+0.5*(45.14-46.11)*0.8*0.8*0.67 = 5.14 kN-m/m
BM_{2x} = (44.74-29.4)*0.8*0.8/2-0.5*(-39.57+44.74)*0.8*0.8*0.67 = 3.81 kN-m/m
SF_{3x} = (46.89-29.4)*(0.8-0.344)+0.5*(45.14-46.89)*(0.8-0.344) = 7.58 kN/m
S.F$_{(punching)}$ = (4.84-0.88)*(42.35-29.4) = 51.28 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	32.00	0.27	0.36	0.001	297.11	619.20	12ϕT	182.65	300	150	753.98
At top	0.00	0.00	0.00	0.000	0.00	309.60	12ϕT	365.30	300	250	452.39

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 43.83 kN/m
Shear stress , τ_v= (43.83*1000)/(1000*344)
= 0.13 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 136.20 kN
Strength of Concrete in shear Vc1 = 3817.91 kN **Table 22.6.5.2_ (a)**
Strength of Concrete in shear Vc2 = 2439.44 kN **Table 22.6.5.2_ (b)**
Strength of Concrete in shear Vc3 = 2545.27 kN **Table 22.6.5.2_ (c)**

Design Vc= 2439.44 kN
Allowable shear stress, (ϕ Vc)/bd = 1.42 N/mm^2

Design punching shear stress, τ = (136.2*1000)/(3752*344)
= 0.11 N/mm^2 **No shear reinforcement reqd.**

DESIGN OF FOUNDATION MKD. F1 of SS#3

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z}	p_{2z}	p_{3z}	$B.M_{1z}$	$B.M_{2z}$	$S.F_{3z}$
	Ultimate Conditions						
301	1.4 DL	55.78	35.07	67.46	14.37	-4.11	21.27
302	1.4 DL + 1.7 LL +0.5LLr	162.27	92.73	201.45	62.40	0.38	91.86
303	1.2 DL+1.6WLx+LL +0.5LLr	124.09	70.91	154.05	45.50	-1.92	67.05
304	1.2 DL+1.6WLz+LL +0.5LLr	117.73	67.27	146.15	42.69	-2.30	62.91
305	1.2 DL+ELx+LL	120.91	69.09	150.10	44.10	-2.11	64.98
306	1.2 DL+ELz+LL	127.27	72.73	158.00	46.91	-1.73	69.12
307	0.9DL+ 1.6WLx	44.47	32.57	51.18	8.23	-2.39	12.18
308	0.9DL+ 1.6WLz	46.23	32.93	53.71	9.18	-2.67	13.60
309	0.9DL+ ELx	48.44	35.11	55.95	9.91	-1.99	14.64
310	0.9DL+ Elz	48.67	34.43	56.69	10.24	-2.46	15.14

Sample calculation of BM & SF

Load case : 301

p_{1z} = 70.67-(70.67-14.04)*0.8/2.2= 50.08 kN/m^2

p_{2z} = 14.04+(70.67-14.04)*0.8/2.2= 34.63 kN/m^2 Overhang = 0.8 m

p_{3z} = 70.67-(70.67-14.04)*(0.8-0.338)/2.2= 58.78 kN/m^2 Effective depth = 0.338 m

BM_{1z} = (55.78-29.4)*0.8*0.8/2+0.5*(70.67-55.78)*0.8*0.8*0.67 = 11.63 kN-m/m

BM_{2z} = (35.07-29.4)*0.8*0.8/2-0.5*(-14.04+35.07)*0.8*0.8*0.67 = -2.69 kN-m/m

SF_{3z} = (67.46-29.4)*(0.8-0.338)+0.5*(70.67-67.46)*(0.8-0.338) = 18.32 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	62.40	0.546	0.728	0.0018	603.97	608.40	12ϕT	185.89	300	150	754.0
At top	-4.11	-0.036	-0.048	-0.0001	-0.39	304.20	12ϕT	371.79	300	250	452.4

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 91.86 kN/m

Shear stress , τ_v= (91.86*1000)/(1000*338)

= 0.27 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	**12ϕT**	**150**	c/c
At Top	**12ϕT**	**250**	c/c
Along Z Direction :			
At Bottom	**12ϕT**	**150**	c/c
At Top	**12ϕT**	**250**	c/c

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DESIGN OF FOUNDATION MKD. F2 of SS#3

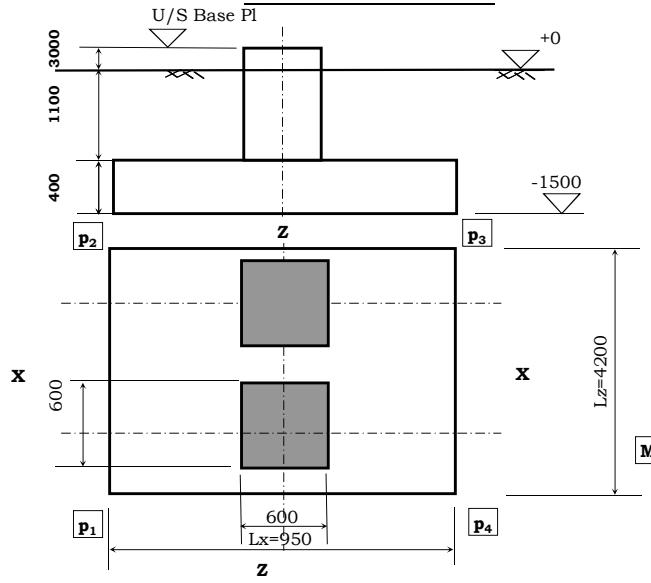
	Along-X	Along-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	0.95m	4.20m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	3.00 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

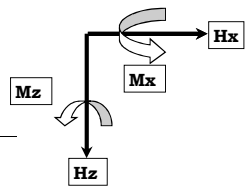
SYMBOLS AND NOTATIONS

Dead Load	DL
Live Load	LL
Wind Load (X)	WLx
Wind Load (Z)	WLz
Earthquake Load (X)	ELx
Earthquake Load (Z)	ELz
Live Load (Roof)	LLr

Calculation of Weight of Foundation :

Weight of raft & Col.=	(0.95*4.2*0.4*24)+(0.6*0.6*4.1*24) =	73.73KN
Weight of soil =	((0.95*4.2)-(0.6*0.6))*1.1*18 =	71.87KN
	Total =	145.60KN





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 344mm = 0.344m

Effective depth of slab along Z direction = 338mm = 0.338m

Clear projection along Z direction = 1.8m

Clear projection along X direction = 0.175m

H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

DESIGN OF FOUNDATION MKD. F2 of SS#3

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)
1	Dead Load (DL)	11.33	-1.68	-0.08	0.47	-0.73
2	Live Load (LL)	10.94	-4.47	-0.10	8.95	-1.37
3	Wind Load (WLx)	17.56	1.25	0.79	-13.96	8.26
4	Wind Load (WLz)	1.44	-2.27	-1.69	-0.90	0.15
5	Earthquake Load (ELx)	-0.03	-2.30	-2.03	0.17	0.00
6	Earthquake Load (ELz)	8.48	0.01	-0.02	-6.05	4.36
7	Live Load (Roof)(LLr)	2.25	-0.40	-0.06	-0.77	-0.24

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	DL+LL	22.27	-6.14	-0.18	9.42	-2.10	167.87	-6.14	-27.83	9.42	40.27
202	DL+0.75LL+0.75LLr	21.23	-5.33	-0.20	6.60	-1.94	166.83	-5.33	-24.17	6.60	27.75
203	DL+WLx	28.89	-0.43	0.71	-13.49	7.53	174.49	-0.43	-1.21	-13.49	-53.20
204	DL+WLz	12.77	-3.95	-1.77	-0.43	-0.58	158.37	-3.95	-19.54	-0.43	-2.52
205	DL+0.7ELx	11.31	-3.29	-1.50	0.59	-0.73	156.91	-3.29	-16.29	0.59	1.90
206	DL+0.7ELz	17.26	-1.67	-0.09	-3.77	2.32	162.86	-1.67	-7.62	-3.77	-14.63
207	DL+0.75WLx+0.75LL+0.75LLr	34.40	-4.39	0.39	-3.87	4.26	180.00	-4.39	-19.36	-3.87	-13.17
208	DL+0.75WLz+0.75LL+0.75LLr	22.31	-7.03	-1.47	5.92	-1.83	167.91	-7.03	-33.11	5.92	24.83
209	DL+0.525ELx+0.75LL+0.75LLr	21.21	-6.54	-1.26	6.69	-1.94	166.81	-6.54	-30.68	6.69	28.15
210	DL+0.525ELz+0.75LL+0.75LLr	25.68	-5.32	-0.21	3.42	0.35	171.28	-5.32	-24.17	3.42	15.75
211	0.6DL + WLx	24.36	0.24	0.74	-13.68	7.82	169.96	0.24	1.84	-13.68	-53.74
212	0.6DL + WLz	8.24	-3.28	-1.74	-0.62	-0.29	153.84	-3.28	-16.49	-0.62	-3.07
213	0.6DL + 0.7ELx	6.78	-2.62	-1.47	0.40	-0.44	152.38	-2.62	-13.24	0.40	1.35
214	0.6DL + 0.7ELx	12.73	-1.00	-0.06	-3.95	2.61	158.33	-1.00	-4.57	-3.95	-15.18

Ultimate Conditions

301	1.4 DL	15.86	-2.35	-0.11	0.65	-1.03	161.46	-2.35	-10.67	0.65	1.92
302	1.4 DL + 1.7 LL +0.5LLr	35.59	-10.14	-0.31	15.48	-3.48	181.19	-10.14	-45.95	15.48	66.19
303	1.2 DL+1.6WLx+LL +0.5LLr	53.77	-4.68	1.03	-13.22	10.85	199.37	-4.68	-20.02	-13.22	-48.63
304	1.2 DL+1.6WLz+LL +0.5LLr	27.97	-10.32	-2.94	7.69	-2.13	173.57	-10.32	-49.36	7.69	32.45
305	1.2 DL+ELx+LL	24.51	-8.78	-2.22	9.68	-2.25	170.11	-8.78	-41.74	9.68	41.31
306	1.2 DL+ELz+LL	33.02	-6.47	-0.21	3.46	2.11	178.62	-6.47	-29.35	3.46	17.69
307	0.9DL+ 1.6WLx	38.30	0.49	1.19	-21.92	12.56	183.90	0.49	3.40	-21.92	-86.07
308	0.9DL+ 1.6WLz	12.50	-5.14	-2.78	-1.02	-0.42	158.10	-5.14	-25.93	-1.02	-5.00
309	0.9DL+ ELx	10.17	-3.81	-2.10	0.59	-0.65	155.77	-3.81	-19.24	0.59	1.99
310	0.9DL+ Elz	18.67	-1.50	-0.09	-5.63	3.70	164.27	-1.50	-6.86	-5.63	-21.62

Check for overturning and sliding:

Load Cases:		Load at B/O Fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	DL+LL	167.87	-6.14	-27.83	9.42	40.27	2.87	1.98	10.93	7.13
202	DL+0.75LL+0.75LLr	166.83	-5.33	-24.17	6.60	27.75	3.28	2.86	12.53	10.11
203	DL+WLx	174.49	-0.43	-1.21	-13.49	-53.20	68.77	1.56	164.04	5.17
204	DL+WLz	158.37	-3.95	-19.54	-0.43	-2.52	3.85	29.80	16.04	146.99
205	DL+0.7ELx	156.91	-3.29	-16.29	0.59	1.90	4.58	39.16	19.09	107.28
206	DL+0.7ELz	162.86	-1.67	-7.62	-3.77	-14.63	10.16	5.29	38.95	17.30
207	DL+0.75WLx+0.75LL+0.75LLr	180.00	-4.39	-19.36	-3.87	-13.17	4.42	6.49	16.40	18.59
208	DL+0.75WLz+0.75LL+0.75LLr	167.91	-7.03	-33.11	5.92	24.83	2.41	3.21	9.55	11.34
209	DL+0.525ELx+0.75LL+0.75LLr	166.81	-6.54	-30.68	6.69	28.15	2.58	2.81	10.21	9.98
210	DL+0.525ELz+0.75LL+0.75LLr	171.28	-5.32	-24.17	3.42	15.75	3.37	5.16	12.87	20.01
211	0.6DL + WLx	169.96	0.24	1.84	-13.68	-53.74	43.82	1.50	277.60	4.97
212	0.6DL + WLz	153.84	-3.28	-16.49	-0.62	-3.07	4.43	23.78	18.77	99.58
213	0.6DL + 0.7ELx	152.38	-2.62	-13.24	0.40	1.35	5.47	53.43	23.29	153.13
214	0.6DL + 0.7ELx	158.33	-1.00	-4.57	-3.95	-15.18	16.46	4.96	63.20	16.02

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DESIGN OF FOUNDATION MKD. F2 of SS#3

Load Combination & Calculation of Base pressure

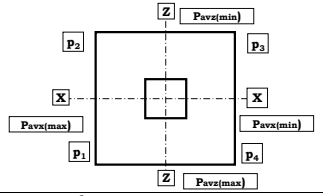
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 500 kN/m^2 For WL/SL case = 625 kN/m^2

Gross SBC of soil for DL, LL= 527 kN/m^2 For WL/SL case = 652 kN/m^2



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)												Remarks
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$P_{avx(max)}$	$P_{avx(min)}$	$P_{avz(max)}$	$P_{avz(min)}$	
201	DL+LL	167.87	27.83	40.27	42.07	44.05	14.42	100.54	71.70	12.44	0.00	73.54	86.12	6.22	56.49	35.85	O.K
202	DL+0.75LL+0.75LLr	166.83	24.17	27.75	41.81	38.26	9.94	90.01	70.14	13.48	0.00	63.01	80.08	6.74	51.75	35.07	O.K
203	DL+WLx	174.49	1.21	53.20	43.73	1.91	19.05	64.69	26.59	60.87	22.78	37.69	45.64	41.83	62.78	24.69	O.K
204	DL+WLz	158.37	19.54	2.52	39.69	30.93	0.90	71.53	69.72	9.67	7.86	44.53	70.63	8.77	40.60	38.79	O.K
205	DL+0.7ELx	156.91	16.29	1.90	39.33	25.79	0.68	65.79	64.43	14.22	12.86	38.79	65.11	13.54	40.01	38.65	O.K
206	DL+0.7ELz	162.86	7.62	14.63	40.82	12.06	5.24	58.11	47.64	34.00	23.52	31.11	52.88	28.76	46.06	35.58	O.K
207	DL+0.75WLx+0.75LL+0.75LL	180.00	19.36	13.17	45.11	30.65	4.72	80.48	71.05	19.18	9.75	53.48	75.77	14.47	49.83	40.40	O.K
208	DL+0.75WLz+0.75LL+0.75LL	167.91	33.11	24.83	42.08	52.42	8.89	103.39	85.61	0.00	0.00	76.39	94.50	0.00	51.70	42.81	O.K
209	DL+0.525ELx+0.75LL+0.75L	166.81	30.68	28.15	41.81	48.56	10.08	100.44	80.29	3.33	0.00	73.44	90.37	1.67	51.89	40.15	O.K
210	DL+0.525ELz+0.75LL+0.75L	171.28	24.17	15.75	42.93	38.26	5.64	86.83	75.55	10.30	0.00	59.83	81.19	5.15	48.57	37.78	O.K
211	0.6DL + WLx	169.96	1.84	53.74	42.60	2.92	19.24	64.76	26.27	58.92	20.44	37.76	45.52	39.68	61.84	23.36	O.K
212	0.6DL + WLz	153.84	16.49	3.07	38.56	26.11	1.10	65.76	63.56	13.55	11.35	38.76	64.66	12.45	39.66	37.46	O.K
213	0.6DL + 0.7ELx	152.38	13.24	1.35	38.19	20.96	0.49	59.64	58.67	17.71	16.74	32.64	59.16	17.23	38.68	37.71	O.K
214	0.6DL + 0.7ELx	158.33	4.57	15.18	39.68	7.23	5.43	52.35	41.48	37.88	27.01	25.35	46.92	32.45	45.12	34.25	O.K
301	1.4 DL	161.46	10.67	1.92	40.47	16.88	0.69	58.04	56.66	24.27	22.90	31.04	57.35	23.59	41.16	39.78	O.K
302	1.4 DL + 1.7 LL +0.5LLr	181.19	45.95	66.19	45.41	72.74	23.70	141.85	94.45	0.00	0.00	114.85	140.00	0.00	70.93	47.23	O.K
303	1.2 DL+1.6WLx+LL +0.5LLr	199.37	20.02	48.63	49.97	31.69	17.41	99.07	64.25	35.68	0.00	72.07	81.66	17.84	67.38	32.13	O.K
304	1.2 DL+1.6WLz+LL +0.5LLr	173.57	49.36	32.45	43.50	78.13	11.62	133.25	110.01	0.00	0.00	106.25	140.00	0.00	66.63	55.01	O.K
305	1.2 DL+ELx+LL	170.11	41.74	41.31	42.64	66.07	14.79	123.49	93.92	0.00	0.00	96.49	130.00	0.00	61.75	46.96	O.K
306	1.2 DL+ELz+LL	178.62	29.35	17.69	44.77	46.46	6.33	97.56	84.89	4.64	0.00	70.56	91.23	2.32	51.10	42.45	O.K
307	0.9DL+ 1.6WLx	183.90	3.40	86.07	46.09	5.39	30.82	82.30	20.66	71.52	0.00	55.30	51.48	35.76	76.91	10.33	O.K
308	0.9DL+ 1.6WLz	158.10	25.93	5.00	39.63	41.05	1.79	82.46	78.88	0.37	0.00	55.46	80.67	0.19	41.42	39.44	O.K
309	0.9DL+ ELx	155.77	19.24	1.99	39.04	30.46	0.71	70.22	68.79	9.29	7.86	43.22	69.51	8.58	39.76	38.33	O.K
310	0.9DL+ Elz	164.27	6.86	21.62	41.17	10.85	7.74	59.76	44.28	38.06	22.58	32.76	52.02	30.32	48.91	33.43	O.K
	Max :	180	33	53													
	Min :	157	1	2													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 73.54 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 83.54 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 63.01 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 73.01 kN/m^2

Sample calculation of values of base pressure:

Load case : DL+0.75LL+0.75LLr

Maximum pressure, p_1 (pmax) = 41.81+38.26+9.94= 90.01 kN/m^2

p_2= 41.81+38.26-9.94= 70.14 kN/m^2

p_3= 41.81-(38.26+9.94)= -6.39 kN/m^2

p_4= 41.81-38.26+9.94= 13.48 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 90.01-1.5*18= 63.01 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (90.01+70.14)/2= 80.07 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (0.00+13.48)/2= 3.55 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (90.01+13.48)/2= 51.75 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (0.00+70.14)/2= 31.87 kN/m^2

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O19045_950_C_K_24310_001_R2	AS BUILT	Rev: D01

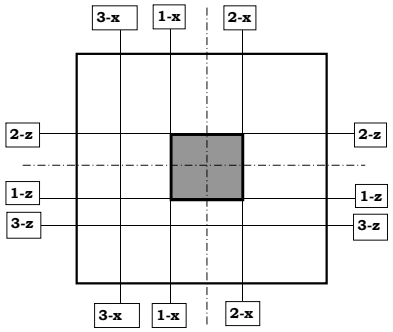
DESIGN OF FOUNDATION MKD. F2 of SS#3

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.
p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.
p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.
BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.
BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.
SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.6+2*0.5*0.338)+(0.6+2*0.5*0.338))*2
= 3.752 m
Punching Area= ((0.6+2*0.5*0.338)*(0.6+2*0.5*0.338))
= 0.88 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.1*18)
= 29.40 kN/m^2
Area of footing, A_f= 3.99 m^2



Load Case	Load Combination	p_{1x}	p_{2x}	p_{3x}	B.M$_{1x}$	B.M$_{2x}$	S.F$_{3x}$	S.F$_{(punching)}$
Ultimate Conditions								
301	1.4 DL	51.13	29.81	63.36	0.40	-0.06	-5.23	34.42
302	1.4 DL + 1.7 LL +0.5LLr	114.21	25.79	164.91	1.56	-0.32	-20.80	49.80
303	1.2 DL+1.6WLx+LL +0.5LLr	69.90	29.60	93.01	0.74	-0.12	-9.79	63.96
304	1.2 DL+1.6WLz+LL +0.5LLr	114.21	25.79	164.91	1.56	-0.32	-20.80	43.86
305	1.2 DL+ELx+LL	106.05	23.95	153.13	1.42	-0.33	-18.96	41.16
306	1.2 DL+ELz+LL	74.85	18.70	107.05	0.86	-0.33	-11.79	47.79
307	0.9DL+ 1.6WLx	48.58	38.66	54.28	0.32	0.11	-3.97	51.91
308	0.9DL+ 1.6WLz	65.84	15.02	94.99	0.71	-0.37	-9.87	31.80
309	0.9DL+ ELx	58.29	19.80	80.35	0.56	-0.26	-7.69	29.98
310	0.9DL+ Elz	48.02	34.32	55.88	0.33	0.03	-4.15	36.61

Sample calculation of BM & SF

Load case : 301

p_{1x} = 52.88-(52.88-28.76)*0.175/0.95 = 48.44 kN/m^2 Overhang = 0.175 m
p_{2x} = 28.76+(52.88-28.76)*0.175/0.95 = 33.20 kN/m^2 Effective depth = 0.344 m
p_{3x} = 52.88-(52.88-28.76)*(0.175-0.344)/0.95 = 57.17 kN/m^2
BM_{1x} = (51.13-29.4)*0.175*0.175/2+0.5*(52.88-51.13)*0.175*0.175*0.67 = 0.35 kN-m/m
BM_{2x} = (29.81-29.4)*0.175*0.175/2-0.5*(-28.76+29.81)*0.175*0.175*0.67 = 0.00 kN-m/m
SF_{3x} = (63.36-29.4)*(0.175-0.344)+0.5*(52.88-63.36)*(0.175-0.344) = -4.85 kN/m
S.F$_{(punching)}$ = (3.99-0.88)*(40.82-29.4) = 35.51 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	1.56	0.01	0.02	0.000	14.42	619.20	12ϕT	182.65	300	150	753.98
At top	-0.37	0.00	0.00	0.000	3.43	309.60	12ϕT	365.30	300	250	452.39

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= -3.97 kN/m
Shear stress , τ_v= (-3.97*1000)/(1000*344)
= -0.01 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 63.96 kN
Strength of Concrete in shear Vc1 = 3817.91 kN **Table 22.6.5.2_ (a)**
Strength of Concrete in shear Vc2 = 2439.44 kN **Table 22.6.5.2_ (b)**
Strength of Concrete in shear Vc3 = 2545.27 kN **Table 22.6.5.2_ (c)**

Design Vc= 2439.44 kN
Allowable shear stress, (ϕ Vc)/bd = 1.42 N/mm^2

Design punching shear stress, τ = (63.96*1000)/(3752*344)
= 0.05 N/mm^2 **No shear reinforcement reqd.**

DESIGN OF FOUNDATION MKD. F2 of SS#3

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z}	p_{2z}	p_{3z}	$B.M_{1z}$	$B.M_{2z}$	$S.F_{3z}$
Ultimate Conditions							
301	1.4 DL	40.57	40.37	40.68	18.74	17.13	16.84
302	1.4 DL + 1.7 LL +0.5LLr	60.77	57.39	62.68	61.85	34.31	54.69
303	1.2 DL+1.6WLx+LL +0.5LLr	52.27	47.24	55.11	53.45	12.50	46.56
304	1.2 DL+1.6WLz+LL +0.5LLr	61.65	59.99	62.59	57.65	44.15	51.47
305	1.2 DL+ELx+LL	55.41	53.30	56.60	49.02	31.84	43.53
306	1.2 DL+ELz+LL	47.39	46.16	48.09	33.17	23.12	29.52
307	0.9DL+ 1.6WLx	48.38	38.86	53.73	61.71	-15.64	52.52
308	0.9DL+ 1.6WLz	40.57	40.29	40.73	19.02	16.72	17.07
309	0.9DL+ ELx	39.15	38.94	39.26	16.46	14.79	14.78
310	0.9DL+ Elz	42.28	40.06	43.52	28.06	10.08	24.58

Sample calculation of BM & SF

Load case : 301

p_{1z} = 46.06-(46.06-35.58)*1.8/4.2= 41.57 kN/m^2

p_{2z} = 35.58+(46.06-35.58)*1.8/4.2= 40.07 kN/m^2 Overhang = 1.8 m

p_{3z} = 46.06-(46.06-35.58)*(1.8-0.338)/4.2= 42.41 kN/m^2 Effective depth = 0.338 m

BM_{1z} = (40.57-29.4)*1.8*1.8/2+0.5*(46.06-40.57)*1.8*1.8*0.67 = 24.05 kN-m/m

BM_{2z} = (40.37-29.4)*1.8*1.8/2-0.5*(-35.58+40.37)*1.8*1.8*0.67 = 12.57 kN-m/m

SF_{3z} = (40.68-29.4)*(1.8-0.338)+0.5*(46.06-40.68)*(1.8-0.338) = 20.42 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	61.85	0.541	0.722	0.0017	598.57	608.40	12ϕT	185.89	300	150	754.0
At top	-15.64	-0.137	-0.183	-0.0004	-1.46	304.20	12ϕT	371.79	300	150	754.0

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 54.69 kN/m

Shear stress , τ_v= (54.69*1000)/(1000*338)

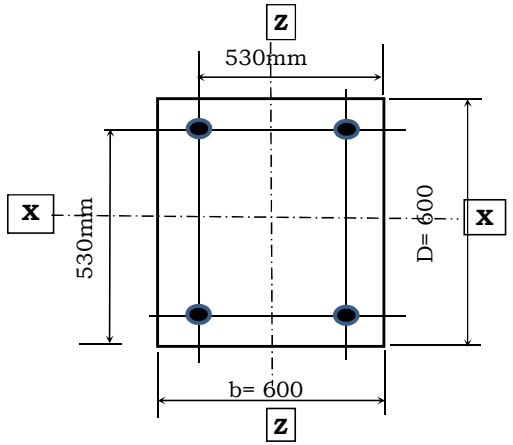
= 0.16 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	**12ϕT**	**150**	c/c
At Top	**12ϕT**	**250**	c/c
Along Z Direction :			
At Bottom	**12ϕT**	**150**	c/c
At Top	**12ϕT**	**150**	c/c

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DESIGN OF COLUMN MKD C1



Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	20mm	
Dia. of link=	10mm	
Effective cover =	70mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	1.00%	
Max. Steel Ratio	4.00%	

For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	4100mm
Unsupproted length of column about Z (l_{uz}) =	4100mm

For Columns are braced against sidesway,

Where, Effective length factor K= 0.8

Slenderness ratio,	$K l_{ux} / r_x$ =	18.94	< 22 SECTION 6.2.5a	Radius of gyration r_x=	173	mm
	$K l_{uz} / r_z$ =	18.94	< 22 SECTION 6.2.5a	Radius of gyration r_z=	173	mm

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Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Dead Load (DL)	35.2	0.7	0.2	10.7	0.2
2	Live Load (LL)	49.5	3.6	-3.6	24.4	0.5
3	Wind Load (WLx)	-9.9	2.6	-1.7	-0.7	0.2
4	Wind Load (WLz)	-6.7	-0.4	-0.3	0.0	0.0
5	Earthquake Load (ELx)	-0.1	-2.7	0.2	0.0	0.0
6	Earthquake Load (ELz)	-1.1	0.7	-0.5	0.7	0.1
7	Live Load (Roof)(LLr)	9.6	-0.8	0.7	0.9	0.0
			γ_y =	0.80		
			γ_z =	0.80		

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f_cbD (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 DL	4.93	1.87	0.03	7.58	2.88	0.04	0.01
302	1.4 DL + 1.7 LL +0.5LLr	13.82	8.41	-2.15	21.26	12.94	3.31	0.02
303	1.2 DL+1.6WLx+LL +0.5LLr	8.07	7.02	-2.23	12.41	10.81	3.44	0.01
304	1.2 DL+1.6WLz+LL +0.5LLr	8.58	5.11	-1.36	13.20	7.87	2.09	0.01
305	1.2 DL+ELx+LL	9.16	4.42	-1.24	14.10	6.80	1.91	0.01
306	1.2 DL+ELz+LL	9.05	5.86	-1.50	13.93	9.01	2.31	0.01
307	0.9DL+ 1.6WLx	1.58	2.82	-0.98	2.43	4.35	1.50	0.00
308	0.9DL+ 1.6WLz	2.10	0.91	-0.10	3.23	1.41	0.16	0.00
309	0.9DL+ ELx	3.16	0.10	0.17	4.86	0.15	0.26	0.00
310	0.9DL+ Elz	3.05	1.54	-0.09	4.69	2.37	0.14	0.00

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Reinforcement: Provide,

4-Nos	20 mm Dia	1257.14 Sq mm (At Corner)	A_{sc} pvd. =	3771.43 Sq mm
4-Nos	20 mm Dia	1257.14 Sq mm (Along Z axis)		
4-Nos	20 mm Dia	1257.14 Sq mm (Along Y axis)	Total A_{sc} =	1.05 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1218.18 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_c b^2 D$	$R_{nx}=M_{nox}/f'_c b^2 D$						
301	0.05	0.05	37.8	37.8	0.55	1.16	0.05	O.K
302	0.05	0.05	37.8	37.8	0.55	1.16	0.35	O.K
303	0.05	0.05	37.8	37.8	0.55	1.16	0.30	O.K
304	0.05	0.05	37.8	37.8	0.55	1.16	0.20	O.K
305	0.05	0.05	37.8	37.8	0.55	1.16	0.17	O.K
306	0.05	0.05	37.8	37.8	0.55	1.16	0.23	O.K
307	0.05	0.05	37.8	37.8	0.55	1.16	0.11	O.K
308	0.05	0.05	37.8	37.8	0.55	1.16	0.02	O.K
309	0.05	0.05	37.8	37.8	0.55	1.16	0.00	O.K
310	0.05	0.05	37.8	37.8	0.55	1.16	0.04	O.K

Links, 10mm dia @ 300 C/C

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DESIGN OF FOUNDATION MKD. F1 of SS#4

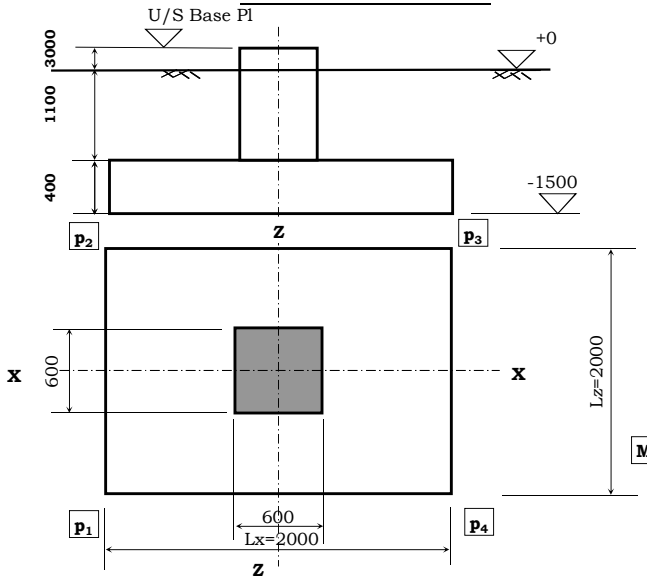
	Along-X	Along-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	2.00m	2.00m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	3.00 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Dead Load	DL
Live Load	LL
Wind Load (X)	WLx
Wind Load (Z)	WLz
Earthquake Load (X)	ELx
Earthquake Load (Z)	ELz
Live Load (Roof)	LLr

Calculation of Weight of Foundation :

Weight of raft & Col.=	(2*2*0.4*24)+(0.6*0.6*4.1*24) =	73.82KN
Weight of soil =	((2*2)-(0.6*0.6))*1.1*18 =	72.07KN
	Total =	145.90KN



Design Input:

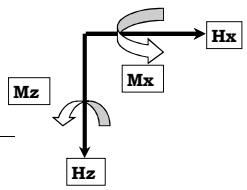
Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 344mm = 0.344m

Effective depth of slab along Z direction = 338mm = 0.338m

Clear projection along Z direction = 0.7m

Clear projection along X direction = 0.7m



H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F1 of SS#4

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)
1	Dead Load (DL)	30.26	-0.70	1.42	11.31	0.26
2	Live Load (LL)	119.43	-0.50	-1.92	9.56	0.10
3	Wind Load (WLx)	-5.14	3.87	-3.68	-14.70	0.11
4	Wind Load (WLz)	-8.66	2.24	-4.24	-0.72	-0.01
5	Earthquake Load (ELx)	0.38	-0.62	0.17	0.03	0.00
6	Earthquake Load (ELz)	0.31	0.40	0.11	-3.89	0.03
7	Live Load (Roof)(LLr)	6.48	-2.39	4.07	0.42	0.01

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	DL+LL	149.69	-1.21	-0.51	20.87	0.36	295.59	-1.21	-5.93	20.87	94.25
202	DL+0.75LL+0.75LLr	124.69	-2.87	3.03	18.79	0.34	270.59	-2.87	-9.88	18.79	84.89
203	DL+WLx	25.12	3.16	-2.26	-3.39	0.37	171.02	3.16	11.97	-3.39	-14.90
204	DL+WLz	21.59	1.54	-2.82	10.59	0.25	167.49	1.54	4.10	10.59	47.89
205	DL+0.7ELx	30.52	-1.14	1.54	11.33	0.26	176.42	-1.14	-3.57	11.33	51.23
206	DL+0.7ELz	30.47	-0.42	1.50	8.58	0.28	176.37	-0.42	-0.40	8.58	38.89
207	DL+0.75WLx+0.75LL+0.75LLr	120.84	0.03	0.27	7.77	0.42	266.74	0.03	0.41	7.77	35.37
208	DL+0.75WLz+0.75LL+0.75LLr	118.19	-1.19	-0.15	18.25	0.33	264.09	-1.19	-5.49	18.25	82.46
209	DL+0.525ELx+0.75LL+0.75LLr	124.89	-3.19	3.12	18.81	0.34	270.79	-3.19	-11.25	18.81	84.97
210	DL+0.525ELz+0.75LL+0.75LLr	124.85	-2.66	3.09	16.75	0.35	270.75	-2.66	-8.87	16.75	75.72
211	0.6DL + WLx	13.02	3.44	-2.83	-7.92	0.26	158.92	3.44	12.67	-7.92	-35.35
212	0.6DL + WLz	9.49	1.82	-3.38	6.06	0.15	155.39	1.82	4.80	6.06	27.44
213	0.6DL + 0.7ELx	18.42	-0.85	0.97	6.81	0.15	164.32	-0.85	-2.87	6.81	30.78
214	0.6DL + 0.7ELx	18.37	-0.14	0.93	4.06	0.18	164.27	-0.14	0.30	4.06	18.44

Ultimate Conditions

301	1.4 DL	42.36	-0.99	1.99	15.83	0.36	188.26	-0.99	-2.45	15.83	71.58
302	1.4 DL + 1.7 LL +0.5LLr	248.63	-3.03	0.75	32.29	0.53	394.53	-3.03	-12.89	32.29	145.84
303	1.2 DL+1.6WLx+LL +0.5LLr	150.76	3.65	-4.07	-0.18	0.59	296.66	3.65	12.34	-0.18	-0.22
304	1.2 DL+1.6WLz+LL +0.5LLr	145.12	1.05	-4.96	22.19	0.40	291.02	1.05	-0.25	22.19	100.24
305	1.2 DL+ELx+LL	156.11	-1.96	-0.05	23.16	0.41	302.01	-1.96	-8.89	23.16	104.63
306	1.2 DL+ELz+LL	156.04	-0.94	-0.11	19.24	0.44	301.94	-0.94	-4.36	19.24	87.00
307	0.9DL+ 1.6WLx	19.01	5.55	-4.61	-13.34	0.41	164.91	5.55	20.38	-13.34	-59.63
308	0.9DL+ 1.6WLz	13.37	2.95	-5.50	9.02	0.22	159.27	2.95	7.79	9.02	40.83
309	0.9DL+ ELx	27.60	-1.25	1.45	10.21	0.23	173.50	-1.25	-4.18	10.21	46.17
310	0.9DL+ Elz	27.53	-0.23	1.39	6.28	0.26	173.43	-0.23	0.35	6.28	28.54

Check for overturning and sliding:

Load Cases:		Load at B/O Fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	DL+LL	295.59	-1.21	-5.93	20.87	94.25	49.82	3.14	98.03	5.67
202	DL+0.75LL+0.75LLr	270.59	-2.87	-9.88	18.79	84.89	27.38	3.19	37.72	5.76
203	DL+WLx	171.02	3.16	11.97	-3.39	-14.90	14.28	11.48	21.63	20.16
204	DL+WLz	167.49	1.54	4.10	10.59	47.89	40.81	3.50	43.56	6.33
205	DL+0.7ELx	176.42	-1.14	-3.57	11.33	51.23	49.39	3.44	62.13	6.23
206	DL+0.7ELz	176.37	-0.42	-0.40	8.58	38.89	440.36	4.53	167.27	8.22
207	DL+0.75WLx+0.75LL+0.75LLr	266.74	0.03	0.41	7.77	35.37	652.98	7.54	3495.90	13.74
208	DL+0.75WLz+0.75LL+0.75LLr	264.09	-1.19	-5.49	18.25	82.46	48.07	3.20	88.93	5.79
209	DL+0.525ELx+0.75LL+0.75LLr	270.79	-3.19	-11.25	18.81	84.97	24.07	3.19	33.92	5.76
210	DL+0.525ELz+0.75LL+0.75LLr	270.75	-2.66	-8.87	16.75	75.72	30.51	3.58	40.75	6.47
211	0.6DL + WLx	158.92	3.44	12.67	-7.92	-35.35	12.54	4.50	18.46	8.03
212	0.6DL + WLz	155.39	1.82	4.80	6.06	27.44	32.35	5.66	34.16	10.25
213	0.6DL + 0.7ELx	164.32	-0.85	-2.87	6.81	30.78	57.21	5.34	76.93	9.66
214	0.6DL + 0.7ELx	164.27	-0.14	0.30	4.06	18.44	549.62	8.91	468.42	16.19

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DESIGN OF FOUNDATION MKD. F1 of SS#4

Load Combination & Calculation of Base pressure

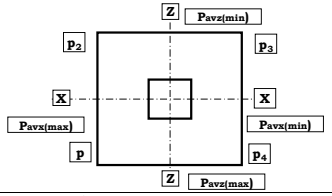
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m²

Net SBC of soil for DL, LL= 500 kN/m² For WL/SL case = 625 kN/m²

Gross SBC of soil for DL, LL= 527 kN/m² For WL/SL case = 652 kN/m²



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m²)													
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p1 (pmax)	p2	p4	p3 (pmin)	(pmax-γh)	Pavx(max)	Pavx(min)	Pavz(max)	Pavz(min)	Remarks
201	DL+LL	295.59	5.93	94.25	73.90	4.45	70.69	149.03	7.66	140.14	0.00	122.03	78.35	70.07	144.59	3.83	O.K
202	DL+0.75LL+0.75LLr	270.59	9.88	84.89	67.65	7.41	63.67	138.73	11.39	123.90	0.00	111.73	75.06	61.95	131.32	5.70	O.K
203	DL+WLx	171.02	11.97	14.90	42.75	8.98	11.18	62.91	40.56	44.95	22.60	35.91	51.74	33.78	53.93	31.58	O.K
204	DL+WLz	167.49	4.10	47.89	41.87	3.08	35.92	80.87	9.03	74.71	2.88	53.87	44.95	38.80	77.79	5.96	O.K
205	DL+0.7ELx	176.42	3.57	51.23	44.10	2.68	38.43	85.21	8.36	79.85	3.00	58.21	46.79	41.43	82.53	5.68	O.K
206	DL+0.7ELz	176.37	0.40	38.89	44.09	0.30	29.17	73.56	15.22	72.96	14.62	46.56	44.39	43.79	73.26	14.92	O.K
207	DL+0.75WLx+0.75LL+0.75LL	266.74	0.41	35.37	66.68	0.31	26.53	93.52	40.46	92.91	39.85	66.52	66.99	66.38	93.22	40.16	O.K
208	DL+0.75WLz+0.75LL+0.75LL	264.09	5.49	82.46	66.02	4.12	61.85	131.99	8.30	123.75	0.05	104.99	70.15	61.90	127.87	4.18	O.K
209	DL+0.525ELx+0.75LL+0.75L	270.79	11.25	84.97	67.70	8.44	63.73	139.86	12.41	122.99	0.00	112.86	76.14	61.50	131.43	6.21	O.K
210	DL+0.525ELz+0.75LL+0.75L	270.75	8.87	75.72	67.69	6.65	56.79	131.13	17.55	117.82	4.24	104.13	74.34	61.03	124.48	10.90	O.K
211	0.6DL + WLx	158.92	12.67	35.35	39.73	9.51	26.52	75.75	22.72	56.74	3.71	48.75	49.24	30.23	66.25	13.22	O.K
212	0.6DL + WLz	155.39	4.80	27.44	38.85	3.60	20.58	63.03	21.87	55.82	14.67	36.03	42.45	35.25	59.43	18.27	O.K
213	0.6DL + 0.7ELx	164.32	2.87	30.78	41.08	2.15	23.09	66.32	20.15	62.01	15.84	39.32	43.24	38.93	64.17	18.00	O.K
214	0.6DL + 0.7ELx	164.27	0.30	18.44	41.07	0.22	13.83	55.12	27.46	54.67	27.01	28.12	41.29	40.84	54.90	27.24	O.K
301	1.4 DL	188.26	2.45	71.58	47.06	1.84	53.69	102.59	0.00	98.91	0.00	75.59	51.30	49.46	100.75	0.00	O.K
302	1.4 DL + 1.7 LL +0.5LLr	394.53	12.89	145.84	98.63	9.67	109.38	217.68	0.00	198.34	0.00	190.68	108.84	99.17	208.01	0.00	O.K
303	1.2 DL+1.6WLx+LL +0.5LLr	296.66	12.34	0.22	74.16	9.26	0.17	83.59	83.25	65.08	64.74	56.59	83.42	64.91	74.34	74.00	O.K
304	1.2 DL+1.6WLz+LL +0.5LLr	291.02	0.25	100.24	72.75	0.19	75.18	148.12	0.00	147.75	0.00	121.12	74.06	73.88	147.94	0.00	O.K
305	1.2 DL+ELx+LL	302.01	8.89	104.63	75.50	6.67	78.47	160.64	3.70	147.31	0.00	133.64	82.17	73.66	153.98	1.85	O.K
306	1.2 DL+ELz+LL	301.94	4.36	87.00	75.49	3.27	65.25	144.00	13.50	137.47	6.97	117.00	78.75	72.22	140.74	10.24	O.K
307	0.9DL+ 1.6WLx	164.91	20.38	59.63	41.23	15.29	44.73	101.24	11.79	70.67	0.00	74.24	56.52	35.34	85.96	5.90	O.K
308	0.9DL+ 1.6WLz	159.27	7.79	40.83	39.82	5.84	30.62	76.28	15.04	64.60	3.35	49.28	45.66	33.98	70.44	9.20	O.K
309	0.9DL+ ELx	173.50	4.18	46.17	43.38	3.13	34.62	81.13	11.89	74.87	5.62	54.13	46.51	40.25	78.00	8.76	O.K
310	0.9DL+ Elz	173.43	0.35	28.54	43.36	0.26	21.40	65.03	22.22	64.50	21.69	38.03	43.63	43.10	64.77	21.96	O.K
	Max :	296	12	94													
	Min :	167	0	15													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 122.03 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 132.03 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 111.73 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 121.73 kN/m²

Sample calculation of values of base pressure:

Load case : DL+0.75LL+0.75LLr

Maximum pressure, p_1 (pmax) = 67.65+7.41+63.67= 138.73 kN/m²

p_2= 67.65+7.41-63.67= 11.39 kN/m²

p_3= 67.65-(7.41+63.67)= 0.00 kN/m²

p_4= 67.65-7.41+63.67= 123.90 kN/m²

Maxm pressure excluding weight of soil (p_{max}-γh)= 138.73-1.5*18= 111.73 kN/m²

Maximum average Pressure along X, $p_{avx(max)}$ = (138.73+11.39)/2= 75.06 kN/m²

Minimum average Pressure along X, $p_{avx(minx)}$ = (0.00+123.90)/2= 61.95 kN/m²

Maximum average Pressure along Y, $p_{avy(max)}$ = (138.73+123.90)/2= 131.32 kN/m²

Minimum average Pressure along Y, $p_{avy(minx)}$ = (0.00+11.39)/2= 5.69 kN/m²

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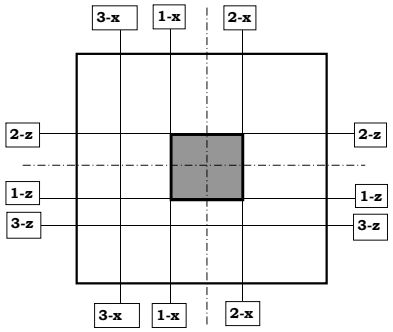
DESIGN OF FOUNDATION MKD. F1 of SS#4

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.
p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.
p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.
BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.
BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.
SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.6+2*0.5*0.338)+(0.6+2*0.5*0.338))*2
= 3.752 m
Punching Area= ((0.6+2*0.5*0.338)*(0.6+2*0.5*0.338))
= 0.88 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.1*18)
= 29.40 kN/m^2
Area of footing, A_f= 4.00 m^2



Load Case	Load Combination	p_{1x}	p_{2x}	p_{3x}	B.M$_{1x}$	B.M$_{2x}$	S.F$_{3x}$	S.F$_{(punching)}$
Ultimate Conditions								
301	1.4 DL	50.66	50.10	50.97	5.31	4.97	7.74	55.11
302	1.4 DL + 1.7 LL +0.5LLr	105.46	102.55	107.12	19.19	17.37	27.97	216.00
303	1.2 DL+1.6WLx+LL +0.5LLr	76.94	71.39	80.13	12.71	9.23	18.64	139.67
304	1.2 DL+1.6WLz+LL +0.5LLr	74.00	73.94	74.03	10.94	10.90	15.89	135.27
305	1.2 DL+ELx+LL	79.19	76.64	80.66	12.69	11.09	18.52	143.84
306	1.2 DL+ELz+LL	76.46	74.51	77.59	11.90	10.68	17.36	143.79
307	0.9DL+ 1.6WLx	49.11	42.75	52.75	6.04	2.06	8.98	36.90
308	0.9DL+ 1.6WLz	41.57	38.07	43.58	3.65	1.46	5.42	32.50
309	0.9DL+ ELx	44.32	42.44	45.40	4.01	2.84	5.89	43.61
310	0.9DL+ Elz	43.44	43.29	43.54	3.47	3.37	5.05	43.55

Sample calculation of BM & SF

Load case : 301

p_{1x} = 44.39-(44.39-43.79)*0.7/2 = 44.18 kN/m^2 Overhang = 0.7 m
p_{2x} = 43.79+(44.39-43.79)*0.7/2 = 44.00 kN/m^2 Effective depth = 0.344 m
p_{3x} = 44.39-(44.39-43.79)*(0.7-0.344)/2 = 44.28 kN/m^2
BM_{1x} = (50.66-29.4)*0.7*0.7/2+0.5*(44.39-50.66)*0.7*0.7*0.67 = 4.18 kN-m/m
BM_{2x} = (50.10-29.4)*0.7*0.7/2-0.5*(-43.79+50.10)*0.7*0.7*0.67 = 4.04 kN-m/m
SF_{3x} = (50.97-29.4)*(0.7-0.344)+0.5*(44.39-50.97)*(0.7-0.344) = 6.51 kN/m
S.F$_{(punching)}$ = (4-0.88)*(44.09-29.4) = 45.84 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	19.19	0.16	0.22	0.001	177.71	619.20	12ϕT	182.65	300	150	753.98
At top	0.00	0.00	0.00	0.000	0.00	309.60	12ϕT	365.30	300	250	452.39

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 27.97 kN/m
Shear stress , τ_v= (27.97*1000)/(1000*344)
= 0.08 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 216.00 kN
Strength of Concrete in shear Vc1 = 3817.91 kN **Table 22.6.5.2_ (a)**
Strength of Concrete in shear Vc2 = 2439.44 kN **Table 22.6.5.2_ (b)**
Strength of Concrete in shear Vc3 = 2545.27 kN **Table 22.6.5.2_ (c)**

Design Vc= 2439.44 kN
Allowable shear stress, (ϕ Vc)/bd = 1.42 N/mm^2

Design punching shear stress, τ = (216*1000)/(3752*344)
= 0.17 N/mm^2 **No shear reinforcement reqd.**

DESIGN OF FOUNDATION MKD. F1 of SS#4

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z}	p_{2z}	p_{3z}	$B.M_{1z}$	$B.M_{2z}$	$S.F_{3z}$
Ultimate Conditions							
301	1.4 DL	65.49	35.26	82.51	14.63	-4.35	22.53
302	1.4 DL + 1.7 LL +0.5LLr	135.21	72.80	170.36	37.87	-1.32	57.84
303	1.2 DL+1.6WLx+LL +0.5LLr	74.22	74.12	74.28	11.00	10.94	16.26
304	1.2 DL+1.6WLz+LL +0.5LLr	96.16	51.78	121.16	24.86	-3.02	38.06
305	1.2 DL+ELx+LL	100.73	55.10	126.44	26.22	-2.44	40.11
306	1.2 DL+ELz+LL	95.07	55.92	117.12	23.59	-1.00	36.03
307	0.9DL+ 1.6WLx	57.94	33.92	71.47	11.59	-3.49	17.85
308	0.9DL+ 1.6WLz	49.01	30.63	59.36	8.32	-3.22	12.85
309	0.9DL+ ELx	53.77	32.99	65.47	9.95	-3.10	15.32
310	0.9DL+ Elz	49.79	36.94	57.02	7.45	-0.61	11.40

Sample calculation of BM & SF

Load case : 301

p_{1z} = 73.26-(73.26-14.92)*0.7/2= 52.84 kN/m^2

p_{2z} = 14.92+(73.26-14.92)*0.7/2= 35.34 kN/m^2 Overhang = 0.7 m

p_{3z} = 73.26-(73.26-14.92)*(0.7-0.338)/2= 62.70 kN/m^2 Effective depth = 0.338 m

BM_{1z} = (65.49-29.4)*0.7*0.7/2+0.5*(73.26-65.49)*0.7*0.7*0.67 = 10.12 kN-m/m

BM_{2z} = (35.26-29.4)*0.7*0.7/2-0.5*(-14.92+35.26)*0.7*0.7*0.67 = -1.90 kN-m/m

SF_{3z} = (82.51-29.4)*(0.7-0.338)+0.5*(73.26-82.51)*(0.7-0.338) = 17.55 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	37.87	0.332	0.442	0.0011	364.76	608.40	12ϕT	185.89	300	150	754.0
At top	-4.35	-0.038	-0.051	-0.0001	-0.41	304.20	12ϕT	371.79	300	250	452.4

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 57.84 kN/m

Shear stress , τ_v= (57.84*1000)/(1000*338)

= 0.17 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **12ϕT** **150** c/c

At Top **12ϕT** **250** c/c

Along Z Direction :

At Bottom **12ϕT** **150** c/c

At Top **12ϕT** **250** c/c

MA'ADEN معادن		Outotec / LARSEN & TOUBRO
MD-216-49PQ-EG-CV-CAL-1625	Mansourah Massarah Gold Project	
900707_GAU01_WD01_CED01_ 45006_OU100684606_R2	DESIGN CALCULATION POWER DISTRIBUTION: PROCESS PLANT SUBSTATIONS FOUNDATION FOR SS – 03 & 04	Date: 25.10.2023
O19045_950_C_K_24310_001_R2	AS BUILT	Rev: D01

DESIGN OF FOUNDATION MKD. F2 of SS#4

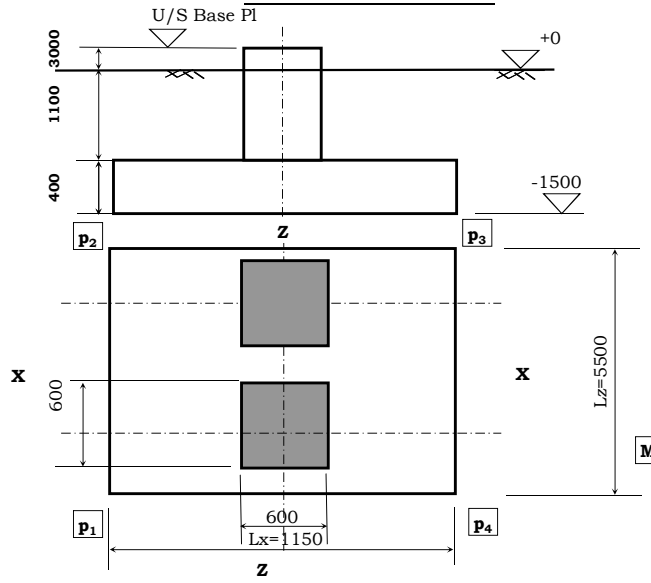
	Along-X	Along-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	1.15m	5.50m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	3.00 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

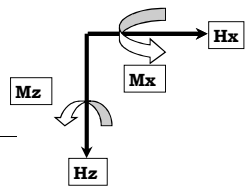
SYMBOLS AND NOTATIONS

Dead Load	DL
Live Load	LL
Wind Load (X)	WLx
Wind Load (Z)	WLz
Earthquake Load (X)	ELx
Earthquake Load (Z)	ELz
Live Load (Roof)	LLr

Calculation of Weight of Foundation :

Weight of raft & Col.=	(1.15*5.5*0.4*24)+(0.6*0.6*4.1*24) =	96.14KN
Weight of soil =	((1.15*5.5)-(0.6*0.6))*1.1*18 =	118.11KN
	Total =	214.25KN





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	16mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 344mm = 0.344m

Effective depth of slab along Z direction = 338mm = 0.338m

Clear projection along Z direction = 2.45m

Clear projection along X direction = 0.275m

H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

DESIGN OF FOUNDATION MKD. F2 of SS#4

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)
1	Dead Load (DL)	11.26	-0.39	0.03	-2.22	-0.82
2	Live Load (LL)	11.89	-0.81	0.14	-2.56	-2.79
3	Wind Load (WLx)	14.41	-1.40	-0.28	-19.07	6.34
4	Wind Load (WLz)	-2.35	-2.99	-2.08	-0.91	0.25
5	Earthquake Load (ELx)	-0.19	-1.67	-1.32	0.07	0.13
6	Earthquake Load (ELz)	5.90	-0.49	-0.14	-6.39	2.68
7	Live Load (Roof)(LLr)	2.71	-0.09	0.04	-0.61	-0.39

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	DL+LL	23.14	-1.20	0.17	-4.78	-3.61	237.39	-1.20	-5.21	-4.78	-25.12
202	DL+0.75LL+0.75LLr	22.20	-1.06	0.17	-4.60	-3.20	236.45	-1.06	-4.61	-4.60	-23.89
203	DL+WLx	25.67	-1.79	-0.24	-21.28	5.52	239.92	-1.79	-8.29	-21.28	-90.26
204	DL+WLz	8.91	-3.38	-2.05	-3.13	-0.57	223.16	-3.38	-17.26	-3.13	-14.65
205	DL+0.7ELx	11.12	-1.56	-0.89	-2.16	-0.73	225.37	-1.56	-7.91	-2.16	-10.46
206	DL+0.7ELz	15.39	-0.73	-0.07	-6.69	1.06	229.64	-0.73	-3.34	-6.69	-29.03
207	DL+0.75WLx+0.75LL+0.75LLr	33.01	-2.11	-0.04	-18.90	1.55	247.26	-2.11	-9.54	-18.90	-83.49
208	DL+0.75WLz+0.75LL+0.75LLr	20.44	-3.31	-1.39	-5.28	-3.01	234.69	-3.31	-16.27	-5.28	-26.78
209	DL+0.525ELx+0.75LL+0.75LLr	22.10	-1.94	-0.52	-4.56	-3.13	236.35	-1.94	-9.26	-4.56	-23.64
210	DL+0.525ELz+0.75LL+0.75LLr	25.30	-1.32	0.10	-7.95	-1.79	239.55	-1.32	-5.83	-7.95	-37.57
211	0.6DL + WLx	21.17	-1.63	-0.26	-20.40	5.85	235.42	-1.63	-7.60	-20.40	-85.94
212	0.6DL + WLz	4.41	-3.22	-2.06	-2.24	-0.24	218.66	-3.22	-16.57	-2.24	-10.33
213	0.6DL + 0.7ELx	6.62	-1.40	-0.90	-1.28	-0.40	220.87	-1.40	-7.22	-1.28	-6.15
214	0.6DL + 0.7ELx	10.88	-0.57	-0.08	-5.80	1.38	225.13	-0.57	-2.66	-5.80	-24.71

Ultimate Conditions

301	1.4 DL	15.76	-0.54	0.05	-3.10	-1.14	230.01	-0.54	-2.40	-3.10	-15.11
302	1.4 DL + 1.7 LL +0.5LLr	37.32	-1.96	0.31	-7.77	-6.08	251.57	-1.96	-8.53	-7.77	-41.03
303	1.2 DL+1.6WLx+LL +0.5LLr	49.81	-3.56	-0.24	-36.04	6.18	264.06	-3.56	-16.26	-36.04	-155.99
304	1.2 DL+1.6WLz+LL +0.5LLr	22.99	-6.11	-3.13	-6.99	-3.56	237.24	-6.11	-30.61	-6.99	-35.01
305	1.2 DL+ELx+LL	25.20	-2.95	-1.14	-5.15	-3.64	239.45	-2.95	-14.41	-5.15	-26.81
306	1.2 DL+ELz+LL	31.29	-1.76	0.04	-11.61	-1.09	245.54	-1.76	-7.88	-11.61	-53.33
307	0.9DL+ 1.6WLx	33.19	-2.59	-0.41	-32.50	9.41	247.44	-2.59	-12.06	-32.50	-136.86
308	0.9DL+ 1.6WLz	6.38	-5.14	-3.30	-3.46	-0.33	220.63	-5.14	-26.42	-3.46	-15.88
309	0.9DL+ ELx	9.94	-2.02	-1.29	-1.92	-0.60	224.19	-2.02	-10.39	-1.92	-9.25
310	0.9DL+ Elz	16.03	-0.83	-0.11	-8.38	1.94	230.28	-0.83	-3.87	-8.38	-35.77

Check for overturning and sliding:

Load Cases:		Load at B/O Fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	DL+LL	237.39	-1.20	-5.21	-4.78	-25.12	26.18	5.43	79.31	19.87
202	DL+0.75LL+0.75LLr	236.45	-1.06	-4.61	-4.60	-23.89	29.49	5.69	89.02	20.58
203	DL+WLx	239.92	-1.79	-8.29	-21.28	-90.26	16.64	1.53	53.67	4.51
204	DL+WLz	223.16	-3.38	-17.26	-3.13	-14.65	7.43	8.76	26.41	28.52
205	DL+0.7ELx	225.37	-1.56	-7.91	-2.16	-10.46	16.39	12.38	57.78	41.66
206	DL+0.7ELz	229.64	-0.73	-3.34	-6.69	-29.03	39.51	4.55	126.15	13.74
207	DL+0.75WLx+0.75LL+0.75LLr	247.26	-2.11	-9.54	-18.90	-83.49	14.90	1.70	46.82	5.23
208	DL+0.75WLz+0.75LL+0.75LLr	234.69	-3.31	-16.27	-5.28	-26.78	8.29	5.04	28.39	17.77
209	DL+0.525ELx+0.75LL+0.75LLr	236.35	-1.94	-9.26	-4.56	-23.64	14.68	5.75	48.70	20.74
210	DL+0.525ELz+0.75LL+0.75LLr	239.55	-1.32	-5.83	-7.95	-37.57	23.62	3.67	72.74	12.05
211	0.6DL + WLx	235.42	-1.63	-7.60	-20.40	-85.94	17.80	1.58	57.67	4.62
212	0.6DL + WLz	218.66	-3.22	-16.57	-2.24	-10.33	7.59	12.17	27.12	38.99
213	0.6DL + 0.7ELx	220.87	-1.40	-7.22	-1.28	-6.15	17.58	20.66	62.89	69.15
214	0.6DL + 0.7ELx	225.13	-0.57	-2.66	-5.80	-24.71	48.72	5.24	157.24	15.53

MD-216-49PQ-EG-CV-CAL-1625	Mansourah Massarah Gold Project	
900707_GAU01_WD01_CED01_ 45006_OU100684606_R2	DESIGN CALCULATION POWER DISTRIBUTION: PROCESS PLANT SUBSTATIONS FOUNDATION FOR SS – 03 & 04	Date: 25.10.2023
O19045_950_C_K_24310_001_R2	AS BUILT	Rev: D01

DESIGN OF FOUNDATION MKD. F2 of SS#4

Load Combination & Calculation of Base pressure

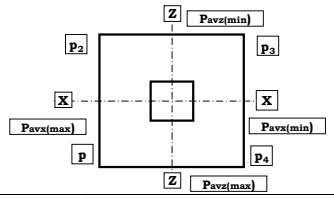
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m²

Net SBC of soil for DL, LL= 500 kN/m² For WL/SL case = 625 kN/m²

Gross SBC of soil for DL, LL= 527 kN/m² For WL/SL case = 652 kN/m²



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m²)													
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p1 (pmax)	p2	p4	p3 (pmin)	(pmax-γh)	pavx(max)	pavx(min)	pavz(max)	pavz(min)	Remarks
201	DL+LL	237.39	5.21	25.12	37.53	4.30	4.33	46.16	37.50	37.56	28.90	19.16	41.83	33.23	41.86	33.20	O.K
202	DL+0.75LL+0.75LLr	236.45	4.61	23.89	37.38	3.80	4.12	45.31	37.07	37.70	29.46	18.31	41.19	33.58	41.51	33.27	O.K
203	DL+WLx	239.92	8.29	90.26	37.93	6.84	15.57	60.34	29.20	46.66	15.53	33.34	44.77	31.10	53.50	22.37	O.K
204	DL+WLz	223.16	17.26	14.65	35.28	14.24	2.53	52.05	46.99	23.57	18.52	25.05	49.52	21.05	37.81	32.76	O.K
205	DL+0.7ELx	225.37	7.91	10.46	35.63	6.52	1.80	43.96	40.35	30.91	27.30	16.96	42.16	29.11	37.44	33.83	O.K
206	DL+0.7ELz	229.64	3.34	29.03	36.31	2.76	5.01	44.07	34.06	38.56	28.54	17.07	39.07	33.55	41.32	31.30	O.K
207	DL+0.75WLx+0.75LL+0.75LL	247.26	9.54	83.49	39.09	7.87	14.40	61.36	32.56	45.62	16.82	34.36	46.96	31.22	53.49	24.69	O.K
208	DL+0.75WLz+0.75LL+0.75LL	234.69	16.27	26.78	37.11	13.42	4.62	55.14	45.91	28.30	19.07	28.14	50.53	23.69	41.72	32.49	O.K
209	DL+0.525ELx+0.75LL+0.75L	236.35	9.26	23.64	37.37	7.64	4.08	49.08	40.93	33.81	25.65	22.08	45.01	29.73	41.45	33.29	O.K
210	DL+0.525ELz+0.75LL+0.75L	239.55	5.83	37.57	37.87	4.81	6.48	49.16	36.20	39.54	26.58	22.16	42.68	33.06	44.35	31.39	O.K
211	0.6DL + WLx	235.42	7.60	85.94	37.22	6.27	14.82	58.31	28.67	45.77	16.13	31.31	43.49	30.95	52.04	22.40	O.K
212	0.6DL + WLz	218.66	16.57	10.33	34.57	13.67	1.78	50.02	46.46	22.68	19.12	23.02	48.24	20.90	36.35	32.79	O.K
213	0.6DL + 0.7ELx	220.87	7.22	6.15	34.92	5.96	1.06	41.94	39.82	30.02	27.90	14.94	40.88	28.96	35.98	33.86	O.K
214	0.6DL + 0.7ELx	225.13	2.66	24.71	35.59	2.19	4.26	42.05	33.52	37.67	29.14	15.05	37.79	33.41	39.86	31.33	O.K
301	1.4 DL	230.01	2.40	15.11	36.37	1.98	2.61	40.95	35.74	36.99	31.78	13.95	38.35	34.39	38.97	33.76	O.K
302	1.4 DL + 1.7 LL +0.5LLr	251.57	8.53	41.03	39.77	7.04	7.08	53.89	39.73	39.81	25.66	26.89	46.81	32.74	46.85	32.70	O.K
303	1.2 DL+1.6WLx+LL +0.5LLr	264.06	16.26	155.99	41.75	13.41	26.90	82.06	28.25	55.24	1.43	55.06	55.16	28.34	68.65	14.84	O.K
304	1.2 DL+1.6WLz+LL +0.5LLr	237.24	30.61	35.01	37.51	25.25	6.04	68.80	56.72	18.30	6.22	41.80	62.76	12.26	43.55	31.47	O.K
305	1.2 DL+ELx+LL	239.45	14.41	26.81	37.86	11.88	4.62	54.37	45.12	30.60	21.35	27.37	49.75	25.98	42.49	33.24	O.K
306	1.2 DL+ELz+LL	245.54	7.88	53.33	38.82	6.50	9.20	54.52	36.13	41.52	23.12	27.52	45.33	32.32	48.02	29.63	O.K
307	0.9DL+ 1.6WLx	247.44	12.06	136.86	39.12	9.95	23.60	72.68	25.47	52.78	5.57	45.68	49.08	29.18	62.73	15.52	O.K
308	0.9DL+ 1.6WLz	220.63	26.42	15.88	34.88	21.79	2.74	59.41	53.93	15.83	10.35	32.41	56.67	13.09	37.62	32.14	O.K
309	0.9DL+ ELx	224.19	10.39	9.25	35.44	8.57	1.59	45.61	42.42	28.47	25.28	18.61	44.02	26.88	37.04	33.85	O.K
310	0.9DL+ Elz	230.28	3.87	35.77	36.41	3.19	6.17	45.77	33.43	39.39	27.05	18.77	39.60	33.22	42.58	30.24	O.K
	Max :	247	17	90													
	Min :	223	3	10													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 19.16 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 29.16 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 33.34 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 43.34 kN/m²

Sample calculation of values of base pressure:

Load case : DL+0.75LL+0.75LLr

Maximum pressure, p_1 (pmax) = 37.38+3.80+4.12= 45.31 kN/m²

p_2= 37.38+3.80-4.12= 37.07 kN/m²

p_3= 37.38-(3.80+4.12)= 29.46 kN/m²

p_4= 37.38-3.80+4.12= 37.70 kN/m²

Maxm pressure excluding weight of soil (p_{max}-γh)= 45.31-1.5*18= 18.31 kN/m²

Maximum average Pressure along X, $p_{avx(max)}$ = (45.31+37.07)/2= 41.19 kN/m²

Minimum average Pressure along X, $p_{avx(minx)}$ = (29.46+37.70)/2= 33.58 kN/m²

Maximum average Pressure along Y, $p_{avy(max)}$ = (45.31+37.70)/2= 41.50 kN/m²

Minimum average Pressure along Y, $p_{avy(minx)}$ = (29.46+37.07)/2= 33.26 kN/m²

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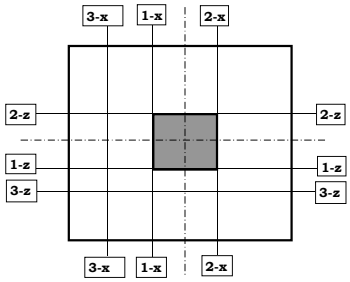
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Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.
p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.
p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.
BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.
BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.
SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.6+2*0.5*0.338)+(0.6+2*0.5*0.338))*2
= 3.752 m
Punching Area= ((0.6+2*0.5*0.338)*(0.6+2*0.5*0.338))
= 0.88 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.1*18)
= 29.40 kN/m^2
Area of footing, A_f= 6.33 m^2



Load Case	Load Combination	p_{1x}	p_{2x}	p_{3x}	$B.M_{1x}$	$B.M_{2x}$	$S.F_{3x}$	$S.F_{(punching)}$
	Ultimate Conditions							
301	1.4 DL	37.40	35.34	38.59	0.33	0.20	-0.63	37.93
302	1.4 DL + 1.7 LL +0.5LLr	43.45	36.10	47.65	0.62	0.17	-1.23	56.49
303	1.2 DL+1.6WLx+LL +0.5LLr	48.75	34.75	56.77	0.89	0.04	-1.83	67.24
304	1.2 DL+1.6WLz+LL +0.5LLr	50.68	24.34	65.79	1.11	-0.50	-2.41	44.15
305	1.2 DL+ELx+LL	44.07	31.66	51.18	0.70	-0.06	-1.45	46.05
306	1.2 DL+ELz+LL	42.22	35.43	46.11	0.56	0.15	-1.13	51.30
307	0.9DL+ 1.6WLx	44.32	33.94	50.27	0.68	0.05	-1.40	52.93
308	0.9DL+ 1.6WLz	46.25	23.51	59.28	0.90	-0.49	-1.97	29.85
309	0.9DL+ ELx	39.92	30.98	45.05	0.50	-0.04	-1.04	32.91
310	0.9DL+ Elz	38.07	34.75	39.98	0.37	0.16	-0.72	38.16

Sample calculation of BM & SF

Load case : 301

p_{1x} = 39.07-(39.07-33.55)*0.275/1.15 = 37.75 kN/m^2 Overhang = 0.275 m
p_{2x} = 33.55+(39.07-33.55)*0.275/1.15 = 34.87 kN/m^2 Effective depth = 0.344 m
p_{3x} = 39.07-(39.07-33.55)*(0.275-0.344)/1.15 = 39.40 kN/m^2
BM_{1x} = (37.40-29.4)*0.275*0.275/2+0.5*(39.07-37.40)*0.275*0.275*0.67 = 0.34 kN-m/m
BM_{2x} = (35.34-29.4)*0.275*0.275/2-0.5*(-33.55+35.34)*0.275*0.275*0.67 = 0.18 kN-m/m
SF_{3x} = (38.59-29.4)*(0.275-0.344)+0.5*(39.07-38.59)*(0.275-0.344) = -0.65 kN/m
$S.F_{(punching)}$ = (6.325-0.88)*(36.31-29.4) = 37.60 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	1.11	0.01	0.01	0.000	10.24	619.20	12ϕT	182.65	300	150	753.98
At top	-0.50	0.00	0.01	0.000	4.58	309.60	12ϕT	365.30	300	250	452.39

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= -0.63 kN/m
Shear stress , τ_v= (-0.63*1000)/(1000*344)
= 0.00 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 67.24 kN
Strength of Concrete in shear Vc1 = 3817.91 kN **Table 22.6.5.2_ (a)**
Strength of Concrete in shear Vc2 = 2439.44 kN **Table 22.6.5.2_ (b)**
Strength of Concrete in shear Vc3 = 2545.27 kN **Table 22.6.5.2_ (c)**

Design Vc= 2439.44 kN
Allowable shear stress, (ϕ Vc)/bd = 1.42 N/mm^2

Design punching shear stress, τ = (67.24*1000)/(3752*344)
= 0.05 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z}	p_{2z}	p_{3z}	$B.M_{1z}$	$B.M_{2z}$	$S.F_{3z}$
	Ultimate Conditions						
301	1.4 DL	36.65	36.08	36.97	26.42	15.38	18.10
302	1.4 DL + 1.7 LL +0.5LLr	40.55	39.00	41.42	46.13	16.15	31.12
303	1.2 DL+1.6WLx+LL +0.5LLr	44.68	38.81	47.99	94.06	-19.96	61.08
304	1.2 DL+1.6WLz+LL +0.5LLr	38.17	36.85	38.91	37.14	11.54	24.99
305	1.2 DL+ELx+LL	38.37	37.36	38.94	35.21	15.61	23.90
306	1.2 DL+ELz+LL	39.83	37.82	40.96	47.77	8.80	31.87
307	0.9DL+ 1.6WLx	41.70	36.55	44.60	79.20	-20.83	51.25
308	0.9DL+ 1.6WLz	35.18	34.58	35.52	22.25	10.64	15.14
309	0.9DL+ ELx	35.62	35.27	35.82	21.52	14.76	14.84
310	0.9DL+ Elz	37.08	35.74	37.84	34.11	7.97	22.83

Sample calculation of BM & SF

Load case : 301

p_{1z} = 41.32-(41.32-31.3)*2.45/5.5= 36.86 kN/m^2

p_{2z} = 31.3+(41.32-31.3)*2.45/5.5= 35.76 kN/m^2

p_{3z} = 41.32-(41.32-31.3)*(2.45-0.338)/5.5= 37.47 kN/m^2

Overhang = 2.45 m

Effective depth = 0.338 m

BM_{1z} = (36.65-29.4)*2.45*2.45/2+0.5*(41.32-36.65)*2.45*2.45*0.67 = 31.15 kN-m/m

BM_{2z} = (36.08-29.4)*2.45*2.45/2-0.5*(-31.3+36.08)*2.45*2.45*0.67 = 10.44 kN-m/m

SF_{3z} = (36.97-29.4)*(2.45-0.338)+0.5*(41.32-36.97)*(2.45-0.338) = 20.58 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	94.06	0.823	1.098	0.0027	916.34	608.40	16ϕT	219.42	300	200	1005.3
At top	-20.83	-0.182	-0.243	-0.0006	-1.95	304.20	16ϕT	660.95	300	200	1005.3

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 61.08 kN/m

Shear stress , τ_v= (61.08*1000)/(1000*338)

= 0.18 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	**12ϕT**	**150**	c/c
	At Top	**12ϕT**	**250**	c/c
Along Z Direction :				
	At Bottom	**16ϕT**	**200**	c/c
	At Top	**16ϕT**	**200**	c/c

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DESIGN OF FOUNDATION MKD. F3 of SS#4

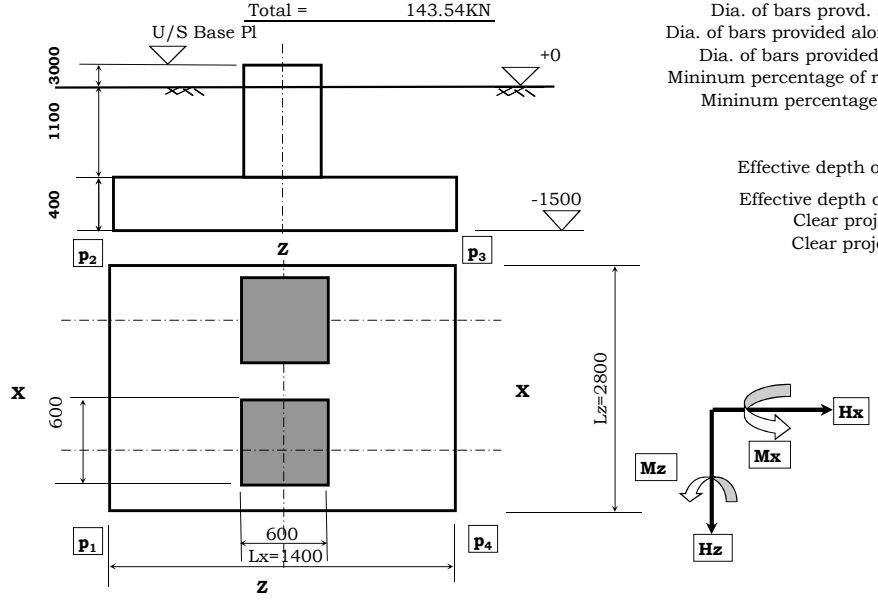
	Along-X	Along-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	1.40m	2.80m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	3.00 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Dead Load	DL
Live Load	LL
Wind Load (X)	WLx
Wind Load (Z)	WLz
Earthquake Load (X)	ELx
Earthquake Load (Z)	ELz
Live Load (Roof)	LLr

Calculation of Weight of Foundation :

Weight of raft & Col.=	(1.4*2.8*0.4*24)+(0.6*0.6*4.1*24) =	73.06KN
Weight of soil =	((1.4*2.8)-(0.6*0.6))*1.1*18 =	70.49KN
	Total =	143.54KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 344mm = 0.344m

Effective depth of slab along Z direction = 338mm = 0.338m

Clear projection along Z direction = 1.1m

Clear projection along X direction = 0.4m

H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)
1	Dead Load (DL)	37.89	-2.66	2.97	-0.33	-4.00
2	Live Load (LL)	60.50	-12.68	6.13	5.08	-6.59
3	Wind Load (WLx)	-2.74	1.41	-2.54	-14.24	8.84
4	Wind Load (WLz)	-6.32	-0.68	-2.50	-2.18	1.70
5	Earthquake Load (ELx)	-0.31	-2.55	-2.61	0.16	0.14
6	Earthquake Load (ELz)	0.33	0.42	-0.83	-3.99	2.86
7	Live Load (Roof)(LLr)	6.75	0.08	0.53	-0.57	-0.26

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	DL+LL	98.39	-15.34	9.09	4.75	-10.59	241.93	-15.34	-59.95	4.75	10.79
202	DL+0.75LL+0.75LLr	88.32	-12.11	7.96	3.05	-9.14	231.86	-12.11	-46.54	3.05	4.60
203	DL+WLx	35.15	-1.26	0.43	-14.58	4.84	178.69	-1.26	-5.22	-14.58	-60.75
204	DL+WLz	31.57	-3.34	0.47	-2.51	-2.30	175.11	-3.34	-14.58	-2.51	-13.60
205	DL+0.7ELx	37.68	-4.45	1.14	-0.22	-3.90	181.22	-4.45	-18.87	-0.22	-4.88
206	DL+0.7ELz	38.12	-2.37	2.38	-3.12	-2.00	181.66	-2.37	-8.26	-3.12	-16.05
207	DL+0.75WLx+0.75LL+0.75LLr	86.27	-11.06	6.06	-7.63	-2.51	229.81	-11.06	-43.70	-7.63	-36.85
208	DL+0.75WLz+0.75LL+0.75LLr	83.59	-12.62	6.09	1.42	-7.86	227.13	-12.62	-50.72	1.42	-1.48
209	DL+0.525ELx+0.75LL+0.75LLr	88.16	-13.45	6.59	3.14	-9.07	231.70	-13.45	-53.93	3.14	5.06
210	DL+0.525ELz+0.75LL+0.75LLr	88.50	-11.89	7.53	0.96	-7.64	232.04	-11.89	-45.98	0.96	-3.32
211	0.6DL + WLx	19.99	-0.19	-0.76	-14.44	6.44	163.53	-0.19	-1.62	-14.44	-58.56
212	0.6DL + WLz	16.42	-2.28	-0.72	-2.38	-0.70	159.96	-2.28	-10.98	-2.38	-11.40
213	0.6DL + 0.7ELx	22.52	-3.38	-0.05	-0.08	-2.30	166.06	-3.38	-15.26	-0.08	-2.68
214	0.6DL + 0.7ELx	22.96	-1.30	1.20	-2.99	-0.40	166.50	-1.30	-4.66	-2.99	-13.85

Ultimate Conditions

301	1.4 DL	53.05	-3.73	4.15	-0.47	-5.60	196.59	-3.73	-12.62	-0.47	-7.69
302	1.4 DL + 1.7 LL +0.5LLr	159.27	-25.25	14.84	7.89	-16.94	302.81	-25.25	-98.77	7.89	18.57
303	1.2 DL+1.6WLx+LL +0.5LLr	104.95	-13.59	5.89	-18.39	2.63	248.49	-13.59	-55.24	-18.39	-80.14
304	1.2 DL+1.6WLz+LL +0.5LLr	99.23	-16.93	5.95	0.91	-8.80	242.77	-16.93	-70.22	0.91	-4.69
305	1.2 DL+ELx+LL	105.66	-18.43	7.08	4.85	-11.26	249.20	-18.43	-75.84	4.85	10.57
306	1.2 DL+ELz+LL	106.29	-15.45	8.86	0.70	-8.53	249.83	-15.45	-60.69	0.70	-5.39
307	0.9DL+ 1.6WLx	29.71	-0.15	-1.39	-23.09	10.55	173.25	-0.15	-2.05	-23.09	-93.36
308	0.9DL+ 1.6WLz	23.99	-3.49	-1.33	-3.79	-0.88	167.53	-3.49	-17.03	-3.79	-17.91
309	0.9DL+ ELx	33.79	-4.94	0.06	-0.13	-3.46	177.33	-4.94	-22.19	-0.13	-4.07
310	0.9DL+ Elz	34.43	-1.97	1.84	-4.28	-0.74	177.97	-1.97	-7.04	-4.28	-20.02

Check for overturning and sliding:

Load Cases:		Load at B/O Fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	DL+LL	241.93	-15.34	-59.95	4.75	10.79	2.82	15.70	6.31	20.37
202	DL+0.75LL+0.75LLr	231.86	-12.11	-46.54	3.05	4.60	3.49	35.31	7.66	30.38
203	DL+WLx	178.69	-1.26	-5.22	-14.58	-60.75	23.95	2.06	56.91	4.90
204	DL+WLz	175.11	-3.34	-14.58	-2.51	-13.60	8.40	9.01	20.94	27.90
205	DL+0.7ELx	181.22	-4.45	-18.87	-0.22	-4.88	6.72	25.99	16.30	333.88
206	DL+0.7ELz	181.66	-2.37	-8.26	-3.12	-16.05	15.39	7.92	30.71	23.28
207	DL+0.75WLx+0.75LL+0.75LLr	229.81	-11.06	-43.70	-7.63	-36.85	3.68	4.37	8.31	12.05
208	DL+0.75WLz+0.75LL+0.75LLr	227.13	-12.62	-50.72	1.42	-1.48	3.13	107.43	7.20	64.04
209	DL+0.525ELx+0.75LL+0.75LLr	231.70	-13.45	-53.93	3.14	5.06	3.01	32.07	6.89	29.53
210	DL+0.525ELz+0.75LL+0.75LLr	232.04	-11.89	-45.98	0.96	-3.32	3.53	48.96	7.81	96.64
211	0.6DL + WLx	163.53	-0.19	-1.62	-14.44	-58.56	70.71	1.95	342.21	4.53
212	0.6DL + WLz	159.96	-2.28	-10.98	-2.38	-11.40	10.20	9.82	28.07	26.90
213	0.6DL + 0.7ELx	166.06	-3.38	-15.26	-0.08	-2.68	7.62	43.32	19.65	788.79
214	0.6DL + 0.7ELx	166.50	-1.30	-4.66	-2.99	-13.85	25.03	8.42	51.19	22.28

MD-216-49PQ-EG-CV-CAL-1625	Mansourah Massarah Gold Project	
900707_GAU01_WD01_CED01_ 45006_OU100684606_R2	DESIGN CALCULATION POWER DISTRIBUTION: PROCESS PLANT SUBSTATIONS FOUNDATION FOR SS – 03 & 04	Date: 25.10.2023
O19045_950_C_K_24310_001_R2	AS BUILT	Rev: D01

DESIGN OF FOUNDATION MKD. F3 of SS#4

Load Combination & Calculation of Base pressure

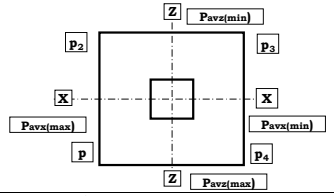
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m²

Net SBC of soil for DL, LL= 500 kN/m² For WL/SL case = 625 kN/m²

Gross SBC of soil for DL, LL= 527 kN/m² For WL/SL case = 652 kN/m²



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m²)													Remarks
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	
201	DL+LL	241.93	59.95	10.79	61.72	65.55	5.90	133.16	121.37	2.07	0.00	106.16	127.27	0.00	67.62	60.69	**O.K**
202	DL+0.75LL+0.75LLr	231.86	46.54	4.60	59.15	50.88	2.51	112.54	107.52	10.78	5.76	85.54	110.03	8.27	61.66	56.64	**O.K**
203	DL+WLx	178.69	5.22	60.75	45.58	5.71	33.21	84.51	18.08	73.08	6.66	57.51	51.30	39.87	78.80	12.37	**O.K**
204	DL+WLz	175.11	14.58	13.60	44.67	15.95	7.43	68.05	53.18	36.16	21.29	41.05	60.62	28.73	52.11	37.24	**O.K**
205	DL+0.7ELx	181.22	18.87	4.88	46.23	20.63	2.67	69.52	64.19	28.27	22.93	42.52	66.86	25.60	48.90	43.56	**O.K**
206	DL+0.7ELz	181.66	8.26	16.05	46.34	9.03	8.77	64.15	46.60	46.08	28.54	37.15	55.38	37.31	55.12	37.57	**O.K**
207	DL+0.75WLx+0.75LL+0.75LL	229.81	43.70	36.85	58.62	47.77	20.14	126.54	86.26	30.99	0.00	99.54	106.40	15.50	78.77	43.13	**O.K**
208	DL+0.75WLz+0.75LL+0.75LL	227.13	50.72	1.48	57.94	55.45	0.81	114.20	112.58	3.30	1.68	87.20	113.39	2.49	58.75	57.13	**O.K**
209	DL+0.525ELx+0.75LL+0.75L	231.70	53.93	5.06	59.11	58.96	2.76	120.83	115.30	2.91	0.00	93.83	118.07	1.46	61.87	57.65	**O.K**
210	DL+0.525ELz+0.75LL+0.75L	232.04	45.98	3.32	59.19	50.27	1.81	111.27	107.64	10.74	7.11	84.27	109.46	8.93	61.01	57.38	**O.K**
211	0.6DL + WLx	163.53	1.62	58.56	41.72	1.77	32.01	75.50	11.48	71.96	7.94	48.50	43.49	39.95	73.73	9.71	**O.K**
212	0.6DL + WLz	159.96	10.98	11.40	40.81	12.00	6.23	59.04	46.58	35.03	22.57	32.04	52.81	28.80	47.04	34.58	**O.K**
213	0.6DL + 0.7ELx	166.06	15.26	2.68	42.36	16.69	1.47	60.51	57.58	27.14	24.21	33.51	59.05	25.68	43.83	40.90	**O.K**
214	0.6DL + 0.7ELx	166.50	4.66	13.85	42.48	5.09	7.57	55.14	40.00	44.95	29.81	28.14	47.57	37.38	50.05	34.91	**O.K**
301	1.4 DL	196.59	12.62	7.69	50.15	13.79	4.21	68.15	59.74	40.56	32.15	41.15	63.95	36.36	54.36	45.95	**O.K**
302	1.4 DL + 1.7 LL +0.5LLr	302.81	98.77	18.57	77.25	107.98	10.15	195.38	175.07	0.00	0.00	168.38	185.23	0.00	97.69	87.54	**O.K**
303	1.2 DL+1.6WLx+LL +0.5LLr	248.49	55.24	80.14	63.39	60.40	43.81	167.60	79.98	46.80	0.00	140.60	123.79	23.40	107.20	39.99	**O.K**
304	1.2 DL+1.6WLz+LL +0.5LLr	242.77	70.22	4.69	61.93	76.77	2.56	141.27	136.14	0.00	0.00	114.27	138.71	0.00	70.64	68.07	**O.K**
305	1.2 DL+ELx+LL	249.20	75.84	10.57	63.57	82.91	5.78	152.26	140.71	0.00	0.00	125.26	146.49	0.00	76.13	70.36	**O.K**
306	1.2 DL+ELz+LL	249.83	60.69	5.39	63.73	66.35	2.94	133.02	127.14	0.33	0.00	106.02	130.08	0.17	66.68	63.57	**O.K**
307	0.9DL+ 1.6WLx	173.25	2.05	93.36	44.20	2.24	51.04	97.47	0.00	92.99	0.00	70.47	48.74	46.50	95.23	0.00	**O.K**
308	0.9DL+ 1.6WLz	167.53	17.03	17.91	42.74	18.62	9.79	71.14	51.56	33.91	14.33	44.14	61.35	24.12	52.53	32.95	**O.K**
309	0.9DL+ ELx	177.33	22.19	4.07	45.24	24.26	2.22	71.72	67.27	23.20	18.75	44.72	69.50	20.98	47.46	43.01	**O.K**
310	0.9DL+ Elz	177.97	7.04	20.02	45.40	7.70	10.94	64.04	42.15	48.65	26.76	37.04	53.10	37.71	56.35	34.46	**O.K**
	Max :	242	60	61														
	Min :	175	5	1														

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 106.16 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 116.16 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 85.54 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 95.54 kN/m²

Sample calculation of values of base pressure:

Load case : DL+0.75LL+0.75LLr

Maximum pressure, p_1 (pmax) = 59.15+50.88+2.51= 112.54 kN/m²

p_2= 59.15+50.88-2.51= 107.52 kN/m²

p_3= 59.15-(50.88+2.51)= 5.76 kN/m²

p_4= 59.15-50.88+2.51= 10.78 kN/m²

Maxm pressure excluding weight of soil (p_{max}-γh)= 112.54-1.5*18= 85.54 kN/m²

Maximum average Pressure along X, $p_{avx(max)}$ = (112.54+107.52)/2 = 110.03 kN/m²

Minimum average Pressure along X, $p_{avx(minx)}$ = (5.76+10.78)/2= 8.27 kN/m²

Maximum average Pressure along Y, $p_{avy(max)}$ = (112.54+10.78)/2= 61.66 kN/m²

Minimum average Pressure along Y, $p_{avy(minx)}$ = (5.76+107.52)/2= 56.64 kN/m²

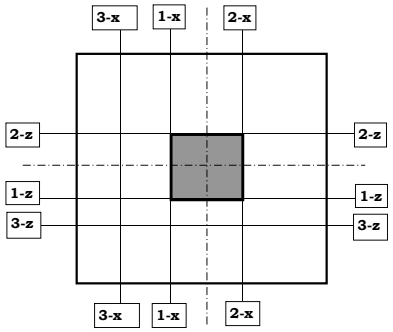
DESIGN OF FOUNDATION MKD. F3 of SS#4

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.
p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.
p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.
BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.
BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.
SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.6+2*0.5*0.338)+(0.6+2*0.5*0.338))*2
= 3.752 m
Punching Area= ((0.6+2*0.5*0.338)*(0.6+2*0.5*0.338))
= 0.88 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.1*18)
= 29.40 kN/m^2
Area of footing, A_f= 3.92 m^2



Load Case	Load Combination	p_{1x}	p_{2x}	p_{3x}	B.M$_{1x}$	B.M$_{2x}$	S.F$_{3x}$	S.F$_{(punching)}$
Ultimate Conditions								
301	1.4 DL	56.07	44.24	62.85	2.55	0.77	1.90	63.08
302	1.4 DL + 1.7 LL +0.5LLr	132.31	52.92	177.82	11.06	-0.94	8.52	145.45
303	1.2 DL+1.6WLx+LL +0.5LLr	95.11	52.08	119.77	6.79	0.28	5.17	103.33
304	1.2 DL+1.6WLz+LL +0.5LLr	99.08	39.63	133.16	7.69	-1.30	5.97	98.90
305	1.2 DL+ELx+LL	104.64	41.85	140.63	8.25	-1.24	6.39	103.88
306	1.2 DL+ELz+LL	92.96	37.29	124.88	7.06	-1.35	5.49	104.37
307	0.9DL+ 1.6WLx	48.10	47.14	48.65	1.53	1.39	1.08	44.98
308	0.9DL+ 1.6WLz	50.71	34.76	59.86	2.27	-0.14	1.75	40.55
309	0.9DL+ ELx	55.64	34.84	67.56	2.84	-0.30	2.19	48.15
310	0.9DL+ Elz	48.70	42.11	52.48	1.78	0.78	1.31	48.64

Sample calculation of BM & SF

Load case : 301

p_{1x} = 55.38-(55.38-37.31)*0.4/1.4 = 50.22 kN/m^2 Overhang = 0.4 m
p_{2x} = 37.31+(55.38-37.31)*0.4/1.4 = 42.47 kN/m^2 Effective depth = 0.344 m
p_{3x} = 55.38-(55.38-37.31)*(0.4-0.344)/1.4 = 54.66 kN/m^2
BM_{1x} = (56.07-29.4)*0.4*0.4/2+0.5*(55.38-56.07)*0.4*0.4*0.67 = 2.10 kN-m/m
BM_{2x} = (44.24-29.4)*0.4*0.4/2-0.5*(-37.31+44.24)*0.4*0.4*0.67 = 0.82 kN-m/m
SF_{3x} = (62.85-29.4)*(0.4-0.344)+0.5*(55.38-62.85)*(0.4-0.344) = 1.66 kN/m
S.F$_{(punching)}$ = (3.92-0.88)*(46.34-29.4) = 51.50 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	11.06	0.09	0.12	0.000	102.24	619.20	12ϕT	182.65	300	150	753.98
At top	-1.35	-0.01	0.02	0.000	12.45	309.60	12ϕT	365.30	300	250	452.39

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 8.52 kN/m
Shear stress , τ_v= (8.52*1000)/(1000*344)
= 0.02 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 145.45 kN
Strength of Concrete in shear Vc1 = 3817.91 kN **Table 22.6.5.2_ (a)**
Strength of Concrete in shear Vc2 = 2439.44 kN **Table 22.6.5.2_ (b)**
Strength of Concrete in shear Vc3 = 2545.27 kN **Table 22.6.5.2_ (c)**

Design Vc= 2439.44 kN
Allowable shear stress, (ϕ Vc)/bd = 1.42 N/mm^2

Design punching shear stress, τ = (145.45*1000)/(3752*344)
= 0.11 N/mm^2 **No shear reinforcement reqd.**

DESIGN OF FOUNDATION MKD. F3 of SS#4

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z}	p_{2z}	p_{3z}	$B.M_{1z}$	$B.M_{2z}$	$S.F_{3z}$
	Ultimate Conditions						
301	1.4 DL	51.06	49.25	52.07	14.44	10.67	18.15
302	1.4 DL + 1.7 LL +0.5LLr	93.70	91.53	94.93	40.52	35.97	50.98
303	1.2 DL+1.6WLx+LL +0.5LLr	80.80	66.39	88.91	41.80	11.68	52.31
304	1.2 DL+1.6WLz+LL +0.5LLr	69.63	69.08	69.94	24.75	23.60	31.16
305	1.2 DL+ELx+LL	73.86	72.63	74.56	27.82	25.23	35.01
306	1.2 DL+ELz+LL	65.46	64.79	65.83	22.31	20.92	28.08
307	0.9DL+ 1.6WLx	57.82	37.41	69.31	32.36	-10.32	40.29
308	0.9DL+ 1.6WLz	44.84	40.64	47.20	12.46	3.68	15.59
309	0.9DL+ ELx	45.71	44.76	46.25	10.58	8.58	13.30
310	0.9DL+ Elz	47.75	43.06	50.39	14.59	4.78	18.27

Sample calculation of BM & SF

Load case : 301

p_{1z} = 55.12-(55.12-37.57)*1.1/2.8= 48.23 kN/m^2

p_{2z} = 37.57+(55.12-37.57)*1.1/2.8= 44.46 kN/m^2 Overhang = 1.1 m

p_{3z} = 55.12-(55.12-37.57)*(1.1-0.338)/2.8= 50.34 kN/m^2 Effective depth = 0.338 m

BM_{1z} = (51.06-29.4)*1.1*1.1/2+0.5*(55.12-51.06)*1.1*1.1*0.67 = 14.75 kN-m/m

BM_{2z} = (49.25-29.4)*1.1*1.1/2-0.5*(-37.57+49.25)*1.1*1.1*0.67 = 7.28 kN-m/m

SF_{3z} = (52.07-29.4)*(1.1-0.338)+0.5*(55.12-52.07)*(1.1-0.338) = 18.44 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	41.80	0.366	0.488	0.0012	402.87	608.40	12ϕT	185.89	300	150	754.0
At top	-10.32	-0.090	-0.120	-0.0003	-0.97	304.20	12ϕT	371.79	300	150	754.0

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 52.31 kN/m

Shear stress , τ_v= (52.31*1000)/(1000*338)

= 0.15 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

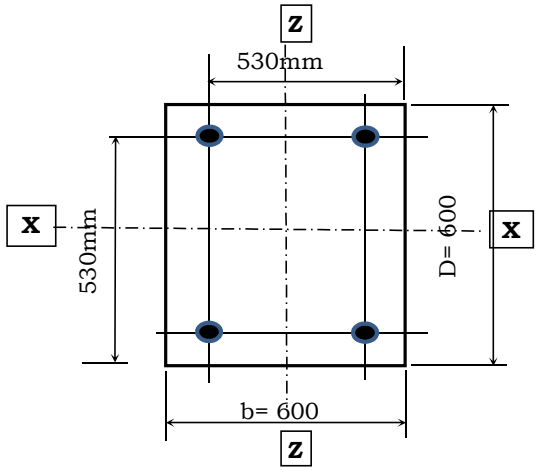
Along X Direction :			
At Bottom	**12ϕT**	**150**	c/c
At Top	**12ϕT**	**250**	c/c
Along Z Direction :			
At Bottom	**12ϕT**	**150**	c/c
At Top	**12ϕT**	**150**	c/c

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MD-216-49PQ-EG-CV-CAL-1625	Mansourah Massarah Gold Project	
900707_GAU01_WD01_CED01_ 45006_OU100684606_R2	DESIGN CALCULATION POWER DISTRIBUTION: PROCESS PLANT SUBSTATIONS FOUNDATION FOR SS – 03 & 04	Date: 25.10.2023
O19045_950_C_K_24310_001_R2	AS BUILT	Rev: D01

DESIGN OF COLUMN MKD C1 of SS#4

Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	20mm	
Dia. of link=	10mm	
Effective cover =	70mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	1.00%	
Max. Steel Ratio	4.00%	



For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	4100mm
Unsupproted length of column about Z (l_{uz}) =	4100mm

For Columns are braced against sidesway,

Where, Effective length factor K= 0.8

Slenderness ratio,	$K\,l_{ux} / r_x$ =	18.94	< 22 SECTION 6.2.5a	Radius of gyration r_x=	173	mm
	$K\,l_{uz} / r_z$ =	18.94	< 22 SECTION 6.2.5a	Radius of gyration r_z=	173	mm

MA'ADEN معادن		Outotec / LARSEN & TOUBRO
MD-216-49PQ-EG-CV-CAL-1625	Mansourah Massarah Gold Project	
900707_GAU01_WD01_CED01_ 45006_OU100684606_R2	DESIGN CALCULATION POWER DISTRIBUTION: PROCESS PLANT SUBSTATIONS FOUNDATION FOR SS – 03 & 04	Date: 25.10.2023
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Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Dead Load (DL)	30.3	-0.7	1.4	11.3	0.3
2	Live Load (LL)	119.4	-0.5	-1.9	9.6	0.1
3	Wind Load (WLx)	-5.1	3.9	-3.7	-14.7	0.1
4	Wind Load (WLz)	-8.7	2.2	-4.2	-0.7	0.0
5	Earthquake Load (ELx)	0.4	-0.6	0.2	0.0	0.0
6	Earthquake Load (ELz)	0.3	0.4	0.1	-3.9	0.0
7	Live Load (Roof)(LLr)	6.5	-2.4	4.1	0.4	0.0
			γ_y =	0.80		
			γ_z =	0.80		

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = $P_n/f'_c bD$ (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 DL	4.24	1.18	0.04	6.52	1.81	0.06	0.01
302	1.4 DL + 1.7 LL +0.5LLr	24.86	1.99	0.36	38.25	3.06	0.56	0.03
303	1.2 DL+1.6WLx+LL +0.5LLr	15.08	1.48	-1.61	23.19	2.27	2.48	0.02
304	1.2 DL+1.6WLz+LL +0.5LLr	14.51	2.65	-1.99	22.33	4.07	3.07	0.02
305	1.2 DL+ELx+LL	15.61	1.51	0.02	24.02	2.32	0.03	0.02
306	1.2 DL+ELz+LL	15.60	1.54	0.00	24.01	2.36	0.00	0.02
307	0.9DL+ 1.6WLx	1.90	0.94	-1.85	2.92	1.45	2.84	0.00
308	0.9DL+ 1.6WLz	1.34	2.11	-2.23	2.06	3.25	3.44	0.00
309	0.9DL+ ELx	2.76	0.51	0.62	4.25	0.78	0.95	0.00
310	0.9DL+ Elz	2.75	0.53	0.60	4.24	0.82	0.92	0.00

MA'ADEN معادن		Outotec / LARSEN & TOUBRO
MD-216-49PQ-EG-CV-CAL-1625	Mansourah Massarah Gold Project	
900707_GAU01_WD01_CED01_ 45006_OU100684606_R2	DESIGN CALCULATION POWER DISTRIBUTION: PROCESS PLANT SUBSTATIONS FOUNDATION FOR SS – 03 & 04	Date: 25.10.2023
O19045_950_C_K_24310_001_R2	AS BUILT	Rev: D01

Reinforcement: Provide,

4-Nos	20 mm Dia	1257.14 Sq mm (At Corner)	A_{sc} pvd. = 3771.43 Sq mm
4-Nos	20 mm Dia	1257.14 Sq mm (Along Z axis)	
4-Nos	20 mm Dia	1257.14 Sq mm (Along Y axis)	Total A_{sc} = 1.05 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc} = 1218.18 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cb^2D$						
301	0.10	0.10	75.6	75.6	0.55	1.16	0.01	O.K
302	0.10	0.10	75.6	75.6	0.55	1.16	0.03	O.K
303	0.10	0.10	75.6	75.6	0.55	1.16	0.04	O.K
304	0.10	0.10	75.6	75.6	0.55	1.16	0.06	O.K
305	0.10	0.10	75.6	75.6	0.55	1.16	0.02	O.K
306	0.10	0.10	75.6	75.6	0.55	1.16	0.02	O.K
307	0.10	0.10	75.6	75.6	0.55	1.16	0.03	O.K
308	0.10	0.10	75.6	75.6	0.55	1.16	0.05	O.K
309	0.10	0.10	75.6	75.6	0.55	1.16	0.01	O.K
310	0.10	0.10	75.6	75.6	0.55	1.16	0.01	O.K

Links, 10mm dia @ 300 C/C

REFERENCES

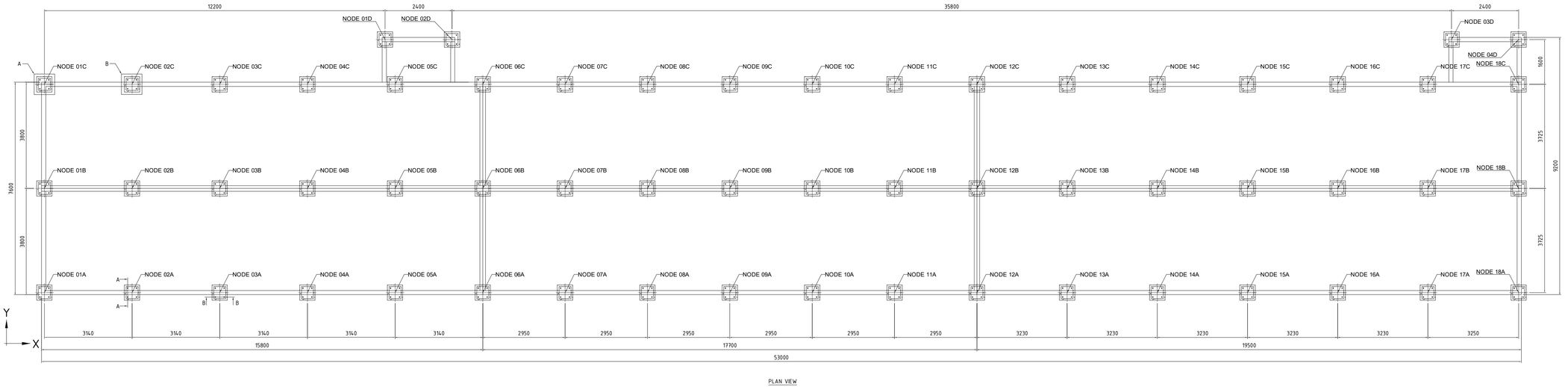
Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

ATTACHMENT 1 LOAD DATA DRAWINGS

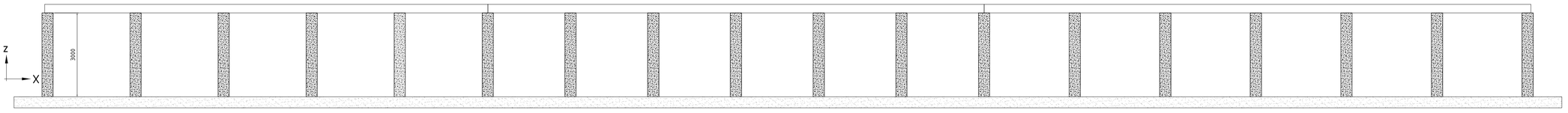
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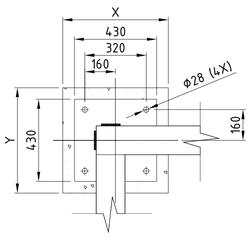
DRAWING NUMBER: MD-216-49PQ-EG-EL-LAD-V552-1001
REV: A01



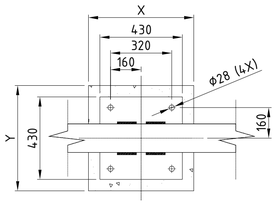
PLAN VIEW



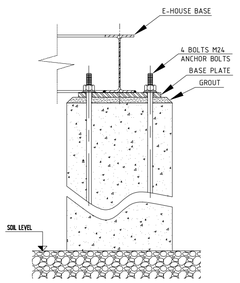
FRONT VIEW



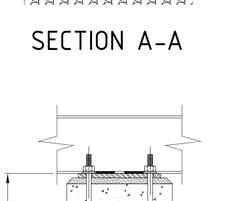
DETAIL A



DETAIL B



SECTION A-A



SECTION B-B

1.0 Dead Load

Node	Fx (lb)	Fy (lb)	Fz (lb)	Mx (lb.m)	My (lb.m)	Mz (lb.m)
Node 01A	1863	1700	13355	4241.1	-3417.7	-83.2
Node 02A	188	7345	26661	9030.2	-3598.2	282.8
Node 03A	211	13173	42205	258.4	-2315.5	422.0
Node 04A	375	8158	32601	178.5	804.9	227.6
Node 05A	661	10659	35182	181.4	355.9	261.6
Node 06A	-1117	3628	27636	7764.1	1006.3	-241.3
Node 07A	-1504	5672	23397	175.6	-1455.8	53.3
Node 08A	-190	5191	25129	182.2	-2.0	-183.9
Node 09A	274	4911	23864	171.4	-664.2	-237.6
Node 10A	194	8185	24723	181.8	330.9	-306.0
Node 11A	560	5269	22058	182.4	1010.8	38.8
Node 12A	1048	1462	21755	4249.8	-508.6	-16.4
Node 13A	-172	984	20389	34.8	185.9	-224.6
Node 14A	-2088	3918	24623	4662.9	-564.9	177.1
Node 15A	-327	5162	27668	206.4	416.1	170.7
Node 16A	537	7169	28604	231.3	704.7	190.7
Node 17A	1610	8402	24209	188.5	1064.1	251.7
Node 18A	-539	642	13190	3673.7	4071.5	103.1
Node 01B	3250	-723	23773	150.2	-4698.3	173.8
Node 02B	-1822	-429	46179	3.9	-955.1	-78.3
Node 03B	2912	-5690	47715	-18.7	-6337.0	-288.8
Node 04B	-1059	-1750	39031	-5.8	864.8	249.3
Node 05B	420	-2414	39187	-10.0	684.7	117.4
Node 06B	-979	-4810	50408	-1550.2	-2934.4	47.9
Node 07B	339	-3920	65482	49.6	-3486.5	-262.9
Node 08B	-383	-5362	73744	68.1	-851.3	-124.8
Node 09B	-170	5635	73020	72.9	2921.6	147.7
Node 10B	875	2229	61318	45.3	-1117.7	261.4
Node 11B	-1515	4722	61247	52.5	10954.0	-148.3
Node 12B	2063	-601	48041	4003.8	-289.7	10.9
Node 13B	-418	9269	47696	120.5	1045.1	-249.3
Node 14B	1304	6578	59211	106.4	-2038.8	23.1
Node 15B	-2123	6704	83174	88.1	9552.1	-31.3
Node 16B	-969	1596	70184	34.4	5616.1	-429.7
Node 17B	-1177	2481	73981	44.7	4443.9	-183.2
Node 18B	-2461	-3229	28269	63.6	7324.1	-18.8
Node 01C	1535	-1914	41035	-4286.9	-3778.9	75.6
Node 02C	345	-7263	21441	-92.6	-336.7	-210.4
Node 03C	906	-6256	23966	-78.4	-1804.6	-262.2
Node 04C	-886	-5777	23193	-76.6	348.2	-166.9
Node 05C	1510	-5722	31916	-82.5	-593.1	-40.0
Node 06C	317	-4291	29183	-1040.9	2965.1	89.2
Node 07C	1208	-8199	21057	-124.5	-1708.7	-346.7
Node 08C	560	-8307	23814	-125.2	-1124.6	-67.8
Node 09C	-228	-8637	24257	-128.1	-619.5	419.0
Node 10C	-362	-11965	24630	-165.7	460.9	84.3
Node 11C	1056	-9233	21629	-131.0	609.0	-310.3
Node 12C	-489	-3180	25467	-1072.4	-1640.4	15.6
Node 13C	0	-11389	28170	-192.2	-509.9	81.1
Node 14C	1207	-10609	28301	417.9	974.1	229.0
Node 15C	375	-9002	29424	-125.1	1102.8	-26.8
Node 16C	-1267	-7492	22607	-91.8	841.3	248.3
Node 17C	-1252	-11284	28523	-144.4	-1400.3	-275.3
Node 18C	-3263	-35	20152	-3237.8	5173.2	-99.4
Node 01D	1181	857	4071	-1257.0	-821.6	479.8
Node 02D	-2857	-189	6661	-612.4	845.1	-268.7
Node 03D	3251	-255	6233	-364.6	-639.7	229.7
Node 04D	-1224	119	6088	-216.9	1004.0	-102.2

1.0 Live Load

Node	Fx (lb)	Fy (lb)	Fz (lb)	Mx (lb.m)	My (lb.m)	Mz (lb.m)
Node 01A	1780	2371	25807	9009.0	-8528.1	-148.1
Node 02A	1734	25271	51099	17268.1	532.2	1205.6
Node 03A	2155	19622	46468	490.2	-631.0	411.0
Node 04A	-167	19277	45164	536.7	393.4	528.5
Node 05A	3063	24268	49478	486.3	-3629.8	772.2
Node 06A	-1253	7241	53962	21438.0	-514.2	-395.8
Node 07A	-1510	12909	41742	417.9	-2411.1	-16.3
Node 08A	249	13109	43236	424.3	-630.8	-483.5
Node 09A	19	19983	43462	468.7	164.6	-542.9
Node 10A	1912	20273	41808	443.0	1962.7	583.5
Node 11A	503	1022	55568	21703.0	-216.9	100.2
Node 12A	-2365	15666	46916	458.5	-13.4	-223.6
Node 13A	-343	13985	47607	477.1	-1307.2	454.7
Node 14A	-390	11014	47721	450.7	1179.2	404.1
Node 15A	-1247	13415	46297	465.6	1373.3	454.7
Node 16A	1610	8402	24209	188.5	1064.1	251.7
Node 17A	-539	642	13190	3673.7	4071.5	103.1
Node 18A	3250	-723	23773	150.2	-4698.3	173.8
Node 01B	-1822	-429	46179	3.9	-955.1	-78.3
Node 02B	2912	-5690	47715	-18.7	-6337.0	-288.8
Node 03B	-1059	-1750	39031	-5.8	864.8	249.3
Node 04B	420	-2414	39187	-10.0	684.7	117.4
Node 05B	-979	-4810	50408	-1550.2	-2934.4	47.9
Node 06B	339	-3920	65482	49.6	-3486.5	-262.9
Node 07B	-383	-5362	73744	68.1	-851.3	-124.8
Node 08B	-170	5635	73020	72.9	2921.6	147.7
Node 09B	875	2229	61318	45.3	-1117.7	261.4
Node 10B	-1515	4722	61247	52.5	10954.0	-148.3
Node 11B	2063	-601	48041	4003.8	-289.7	10.9
Node 12B	-418	9269	47696	120.5	1045.1	-249.3
Node 13B	1304	6578	59211	106.4	-2038.8	23.1
Node 14B	-2123	6704	83174	88.1	9552.1	-31.3
Node 15B	-969	1596	70184	34.4	5616.1	-429.7
Node 16B	-1177	2481	73981	44.7	4443.9	-183.2
Node 17B	-2461	-3229	28269	63.6	7324.1	-18.8
Node 18B	1535	-1914	41035	-4286.9	-3778.9	75.6
Node 01C	345	-7263	21441	-92.6	-336.7	-210.4
Node 02C	906	-6256	23966	-78.4	-1804.6	-262.2
Node 03C	-886	-5777	23193	-76.6	348.2	-166.9
Node 04C	1510	-5722	31916	-82.5	-593.1	-40.0
Node 05C	317	-4291	29183	-1040.9	2965.1	89.2
Node 06C	1208	-8199	21057	-124.5	-1708.7	-346.7
Node 07C	560	-8307	23814	-125.2	-1124.6	-67.8
Node 08C	-228	-8637	24257	-128.1	-619.5	419.0
Node 09C	-362	-11965	24630	-165.7	460.9	84.3
Node 10C	1056	-9233	21629	-131.0	609.0	-310.3
Node 11C	-489	-3180	25467	-1072.4	-1640.4	15.6
Node 12C	0	-11389	28170	-192.2	-509.9	81.1
Node 13C	1207	-10609	28301	417.9	974.1	229.0
Node 14C	375	-9002	29424	-125.1	1102.8	-26.8
Node 15C	-1267	-7492	22607	-91.8	841.3	248.3
Node 16C	-1252	-11284	28523	-144.4	-1400.3	-275.3
Node 17C	-3263	-35	20152	-3237.8	5173.2	-99.4
Node 18C	1181	857	4071	-1257.0	-821.6	479.8
Node 01D	-2857	-189	6661	-612.4	845.1	-268.7
Node 02D	3251	-255	6233	-364.6	-639.7	229.7
Node 03D	-1224	119	6088	-216.9	1004.0	-102.2
Node 04D	1734	25271	51099	17268.1	532.2	1205.6

1.0 Wind Load Transversal

Node	Fx (lb)	Fy (lb)	Fz (lb)	Mx (lb.m)	My (lb.m)	Mz (lb.m)
Node 01A	1000	-8511	-9618	3271.7	2537.1	-342.7
Node 02A	2133	-394	-14414	1463.2	1329.8	-1121.3
Node 03A	1119	-219	-9149	160.5	270.6	-1144.1
Node 04A	357	-1122	-11617	170.2	599.3	-588.9
Node 05A	-353	-4573	-9955	157.3	-1698.1	-56.8
Node 06A	-1350	-8352	-11976	4919.7	-609.4	1082.4
Node 07A	-1734	-5047	-10319	137.2	-681.1	-254.4
Node 08A	-878	-1852	-10104	204.5	38.2	137.5
Node 09A	-149	290	-10074	243.6	3.0	934.4
Node 10A	667	762	-9973	230.8	-254.1	-1158.5
Node 11A	2202	-1595	-10565	168.8	1704.4	71.9
Node 12A	-691	-12261	-13231	3649.9	236.8	1261.6
Node 13A	-3487	3639	-12604	267.4	-227.0	1898.6
Node 14A	-424	-11270	-10441	5036.7	-232.8	1844.5
Node 15A	-480	-3296	-11166	243.2	-424.4	-218.7
Node 16A	903	-6009	-10473	198.4	-186.5	206.1
Node 17A	100	-3447	-9673	154.1	-1214.1	-263.7
Node 18A	-91	-6947	-9214	2684.3	-2226.4	389.3
Node 01B	954	-11025	-4399	-665.7	828.0	0.0
Node 02B	2251	-16007	-6408	-12.4	659.3	28.5
Node 03B	-2138	-18989	-11027	-33.5	5183.9	-668.6
Node 04B	960	-16199	-847	-340.4	-3.3	-2516.3
Node 05B	62	-15548	-948	-9.5	-11.3	-243.1
Node 06B	1504	-18843	-13575	416.3	2231.4	-39.5
Node 07B	-1037	-11314	-5878	13.1	2507.9	-398.9
Node 08B	541	-10118	-4380	-24.6	-1208.1	-65.4
Node 09B	360	-13620	-11471	42.2	-3908.9	214.0
Node 10B	-1036	-20300	-1	-7.5	746.2	654.0
Node 11B	-3948	-12912	-6550	31.3	-6426.0	-917.2
Node 12B	-2078	-23950	-14884	1988.2	-1652.2	70.4
Node 13B	-142	-17395	1424	-1.7	-296.0	631.4
Node 14B	-1396	-10783	-785	27.0	1479.7	285.0
Node 15B	1719	-2334	-12161	140.6	-7264.2	746.0
Node 16B	-342	-12278	4774	38.4	-2732.9	-186.5
Node 17B	-1778	-5490	-6811	38.2	-1244.0	-1537.8
Node 18B	-1802	-10022	-3644	350.1	-833.3	22.8
Node 01C	2068	-7825	5046	5741.4	328.9	-1215.3
Node 02C	565	1915	364	144.4	-194.6	-887.9
Node 03C	272	2139	414	156.7	-107.9	-1089.8
Node 04C	-250	876	-1880	149.7	1185.0	-679.4
Node 05C	-2312	778	-13225	62.6	1706.6	28.3
Node 06C	2643	-6191	-3088	5099.0	-2501.1	1180.2
Node 07C	1395	-4037	1866	-1258.4	3702.8	-155.5
Node 08C	1008	-3128	1579	170.8	86.6	-161.3
Node 09C	-37	-1234	1183	199.4	238.4	515.1
Node 10C	-711	-183	764	190.1	2577.3	-337.5
Node 11C	-2400	-634	2186	156.6	-673.8	-198.0
Node 12C	270	-9135	4184	4467.9	439.3	856.5
Node 13C	1671	1780	-425	195.9	795.9	1236.9
Node 14C	1412	-7606	-1704	3604.3	-446.7	241.1
Node 15C	179	-425	1166	244.6	-50.9	-241.9
Node 16C	-1892	-3680	1101	161.5	64.4	-23.3
Node 17C	-4640	-1488	-5399	131.3	2457.6	208.6
Node 18C	-695	-8241	-851	3607.0	-2855.9	337.2
Node 01D	-480	8355	3145	-144.1	-1572.9	-138.4
Node 02D	-3897	2595	7798	-1515.7	1472.6	-465.9
Node 03D	6892	928	8332	-1399.2	-1863.9	335.6
Node 04D	-1135	552	7471	-842.6	2021.6	84.6

1.0 Wind Load Longitudinal

Node	Fx (lb)	Fy (lb)	Fz (lb)	Mx (lb.m)	My (lb.m)	Mz (lb.m)
Node 01A	-3726	894	-4367	-2285.1	-403.3	125.6
Node 02A	-3723	520	-4403	-1417.7	-1296.3	21.9
Node 03A	-957	-351	-5213	-24.0	92.3	50.3
Node 04A	-1088	-147	-6529	-15.8	398.2	47.3
Node 05A	-437	-9	-6674	-11.7	-296.3	21.9
Node 06A	-381	215	-5432	-2023.2	-292.5	-52.8
Node 07A	-919	198	-4377	-24.3	302.1	48.9
Node 08A	-545	-280	-4792	-27.1	79.4	27.2
Node 09A	-882	-362	-4993	-25.2	199.9	-9.9
Node 10A	-689	-316	-3808	-23.5	-367.2	-67.3
Node 11A	-707	-195	-3265	-22.8	-97.1	23.5
Node 12A	-929	800	-2936	-1101.8	294.5	-107.1
Node 13A	-635	-49	-3682	-15.5	257.9	-35.4
Node 14A	-1274	-1241	-3826	-341.1	-108.1	-3.0
Node 15A	-781	-312	-2779	-29.9	-155.3	11.7
Node 16A	-1134	52	-2527	-22.2	-12.0	-10.9
Node 17A	-1247	-1179	-2714	-28.4	-14.0	-15.7
Node 18A	-2387	2036	-412	-643.5	-1347.0	20.1
Node 01B	-4242	-254	-7415	-1189.5	-1866.2	-195.8
Node 02B	-9507	-6482	-12414	859.3	28.5	-1.4
Node 03B	-4266	-778	-15805	-5.4	6202.2	-32.2
Node 04B	-1064	-300	-4704	-3.3	-2516.3	40.0
Node 05B	-244	-216	-948	-9.5	-11.3	-243.1
Node 06B	-1578	791	-18937	-341.2	1521.9	0.4
Node 07B	-1008	-545	-9630	-8.1	2788.8	24.0
Node 08B	-253	-552	-1608	-9.0	-639.5	22.5
Node 09B	423	-1235	-14211	-13.9	-4087.4	-4.9
Node 10B	940	-91	-586	-0.3	401.9	-3.4
Node 11B	1423	-440	-4951	-4.8	-4146.2	12.7
Node 12B	-2036	-98	-10598	-18.5	-1435.6	-1.5
Node 13B	-180	-145	-621	-0.9	-191.1	5.3
Node 14B	-1147	-183	-2376	-2.9	1092.2	-13.4
Node 15B	854	-1080	-9979	-12.4	-5572.6	21.1
Node 16B	-458	-513	-6372	-7.1	-1896.5	38.7
Node 17B	-1778	-5490	-6811	38.2	-1244.0	-1537.8
Node 18B	-1913	878	-2736	-68.1	-1513.0	-112.1
Node 01C	-4097	-875	-4476	2447.4	-798.1	-131.9
Node 02C	-2222	-994	-4168	17.2	250.2	-68.5
Node 03C	-1332	919	-5155	26.3	72.0	-17.2
Node 04C	-1077	511	-6428	16.7	1185.0	-679.4
Node 05C	-717	330	-7710	9.3	-342.2	-43.9
Node 06C	-473	186	-5984	1750.4	-1218.2	69.6
Node 07C	1128	261	-4467	19.4	370.8	-15.5
Node 08C	-762	822	-4702	20.0	226.8	-17.4
Node 09C	-822	869	-4877	19.5	226.4	-21.4
Node 10C	-2577	-21	-10	0.6	188.3	-16.7
Node 11C	-2562	64	-112	0.8	274.5	-19.2
Node 12C	-2466	-67	103	-48.2	-57.4	-24.9
Node 13C	-1050	211	-3541	19.5	287.1	66.2
Node 14C	-869	-692	-3152	433.8	-388.5	5.4
Node 15C	-1058	1068	-2694	30.0	-150.4	-37.0
Node 16C	-2706	17	-4	-0.5	109.2	-19.1
Node 17C	-1626	421	-2790	20.8	146.9	14.7
Node 18C	-1829	-1429	-402	314.4	-1663.2	36.2
Node 01D	-1834	-478	606	105.5	-1219.4	-108.4
Node 02D	-339	-223	836	42.1	-475.5	29.4
Node 03D	-321	-115	-273	17.7	-514.9	22.9
Node 04D	-1954	-205	555	-71.3	-833.9	-81.9

1.0 Seismic Load X

Node	Fx (lb)	Fy (lb)	Fz (lb)	Mx (lb.m)	My (lb.m)	Mz (lb.m)
Node 01A	-2130	59	-2974	-604.1	-1330.6	87.2
Node 02A	-2636	-90	-52	0.0	75.2	19.1
Node 03A	-2644	-55	-26	0.0	140.2	19.7
Node 04A	-2664	11	-74	-0.1	187.9	14.8
Node 05A	-2640	31	8	136.9	-384.8	39.2
Node 06A	-2556	82	121	1.1	89.8	12.7
Node 07A	-2505	36	-44	-0.6	83.1	5.7
Node 08A	-2498	4	-37	-1.1	151.7	21.9
Node 09A	-2521	27	-24	-0.5	172.0	16.3
Node 10A	-2547	-42	-60	-1.0	256.4	20.1
Node 11A	-2579	87	107	48.3	-32.4	36.4
Node 12A	-2604	118	207	0.5	-25.4	5.8
Node 13A	-2634	36	-82	-4.6	-362.5	-1.5
Node 14A	-2684	-75	-108	-1.3	-24.0	16.5
Node 15A	-2716	25	52	-0.6	87.3	19.0
Node 16A	-2633	-156	-428	-4.5	348.7	14.0
Node 17A	-2203	87	2961	666.0	-1359.5	76.6
Node 18A	-1035	-58	-1567	14.7	-1376.0	-188.8
Node 01B	-1918	-140	-2048	107.9	-532.2	81.3
Node 02B	-958	-6	-375	-0.1	50.0	3.1
Node 03B	-948	69	618	0.6	277.5	-2.2
Node 04B	-944	21	1103	2.3	1746.0	-14.2
Node 05B	-1383	9	-398	308.8	-2031.4	-4.0
Node 06B	-1224	-119	-1137	2.3	-282.0	11.3
Node 07B	-1264	-116	-2082	-0.3	1746.0	-14.2
Node 08B	-1018	101	2042	1.9	-273.8	15.5
Node 09B	-940	-61	-586	-0.3	401.9	-3.4
Node 10B	-1423	-440	-4951	-4.8	-4146.2	12.7
Node 11B	-2087	20	1459	4.8	-1428.7	5.5
Node 12B	-1482	34	804	2.2	122.1	-0.1
Node 13B	-956	-56	-788	-0.8	420.5	-4.7
Node 14B	-1380	-979	-12	-1.8	-2230.3	7.9
Node 15B	-958	-513	-6372	-7.1	-1896.5	38.7
Node 16B	-1781	-153	-1292	-1.1	-276.7	9.0
Node 17B	-1913	878	-2736	-68.1	-1513.0	-112.1
Node 01C	-4097	-875	-4476	2447.4	-798.1	-131.9
Node 02C	-2222	-994	-4168	17.2	250.2	-68.5
Node 03C	-1332	919	-5155	26.3	72.0	-17.2
Node 04C	-1077	511	-6428	16.7	1185.0	-679.4
Node 05C	-717	330	-7710	9.3	-342.2	-43.9
Node 06C	-473	186	-5984	1750.4	-1218.2	69.6
Node 07C	1128	261	-4467	19.4	370.8	-15.5
Node 08C	-762	822	-4702	20.0	226.8	-17.4
Node 09C	-822	869	-4877	19.5	226.4	-21.4
Node 10C	-2577	-21	-10	0.6	188.3	-16.7
Node 11C	-2562	64	-112	0.8	274.5	-19.2
Node 12C	-2466	-67	103	-48.2	-57.4	-24.9
Node 13C	-1050	211	-3541	19.5	287.1	66.2
Node 14C	-869	-692	-3152	433.8	-388.5	5.4
Node 15C	-1058	1068	-2694	30.0	-150.4	-37.0
Node 16C	-2706	17	-4	-0.5	109.2	-19.1
Node 17C	-1626	421	-2790	20.8	146.9	14.7
Node 18C	-1829	-1429	-402	314.4	-1663.2	36.2
Node 01D	-1834	-478	606	105.5	-1219.4	-108.4
Node 02D	-339	-223	836	42.1	-475.5	29.4
Node 03D	-321	-115	-273	17.7	-514.9	22.9
Node 04D	-1954	-205	555	-71.3	-833.9	-81.9

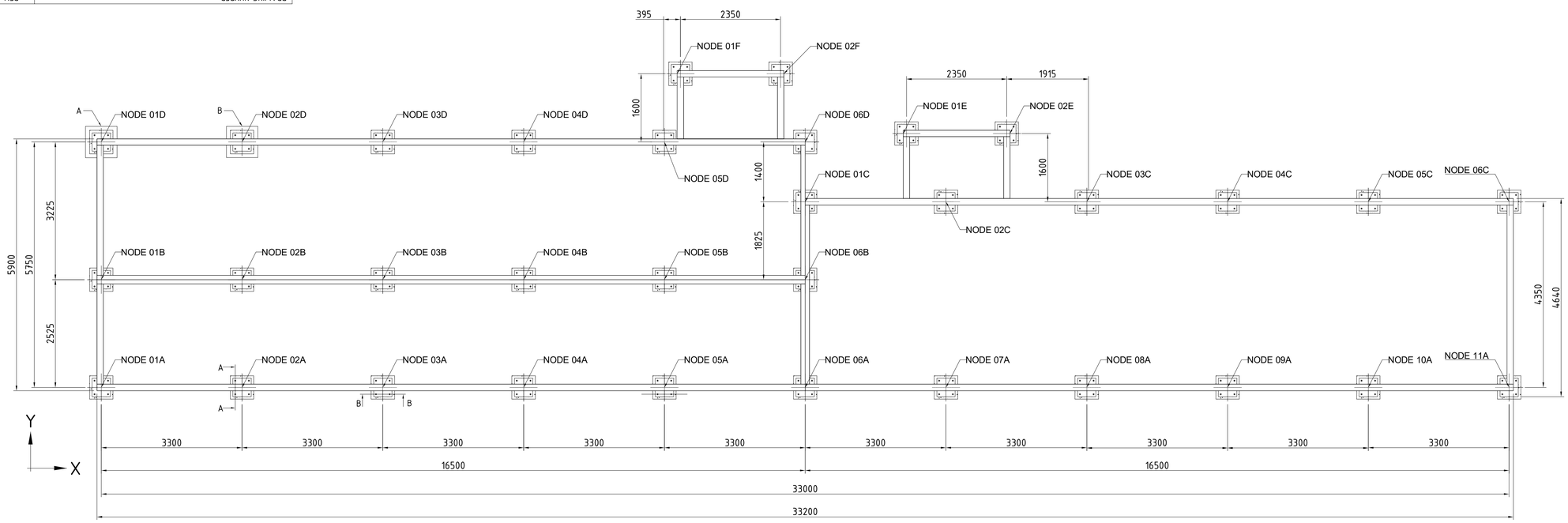
1.0 Seismic Load Y

Node	Fx (lb)	Fy (lb)	Fz (lb)	Mx (lb.m)	My (lb.m)	Mz (lb.m)
Node 01A	6	-2791	-3311	1670.2	423.9	-123.8
Node 02A	179	786	-1574	56.6	65.2	-376.4
Node 03A	194	549	-1496	54.1	89.2	-175.1
Node 04A	661	655	-1147	53.7	-458.6	21.3
Node 05A	-636	-1641	-3063	2438.2	-122.5	441.9
Node 06A	-424	-1249	-2147	48.3	-290.4	-73.6
Node 07A	-238	-1109	-2025	69.9	-16.3	53.5
Node 08A	-63	669	-1875	84.2	-62.6	332.1
Node 09A	229	1081	-1704	81.8	-75.2	398.3
Node 10A	1059	-232	-2222	82.0	188.7	31.9
Node 11A	-143	-3462	-3476	1633.0	-129.2	487.9
Node 12A	144	-2040	-1902	89.4	-250.0	615.2
Node 13A	-2934	-2837	-3278	1830.8	81.3	54.7
Node 14A	-100	-895	-2087	88.0	-52.5	-90.8
Node 15A	691	-1730	-1899	57.9	-93.7	74.8
Node 16A	-1033	-1156	-428	-4.5	348.7	14.0
Node 17A	-2203	87	2961	666.0	-1359.5	76.6
Node 18A	-1035	-58	-1567	14.7	-1376.0	-188.8
Node 01B	-1918	-140	-2048	107.9	-532.2	81.3
Node 02B	-958	-6	-375	-0.1	50.0	3.1
Node 03B	-948	69	618	0.6	277.5	-2.2
Node 04B	-944	21	1103	2.3	1746.0	-14.2
Node 05B	-1383	9	-398	308.8	-2031.4	-4.0
Node 06B	-1224	-119	-1137	2.3	-282.0	11.3
Node 07B	-1264	-116	-2082	-0.3	1746.0	-14.2
Node 08B	-1018	101	2042	1.9	-273.8	15.5
Node 09B	-940	-61	-586	-0.3	401.9	-3.4
Node 10B	-1423	-440	-4951	-4.8	-4146.2	12.7
Node 11B	-2087	20	1459	4.8	-1428.7	5.5
Node 12B	-1482	34	804	2.2	122.1	-0.1
Node 13B	-956	-56	-788	-0.8	420.5	-4.7
Node 14B	-1380	-979	-12	-1.8	-2230.3	7.9
Node 15B	-958	-513	-6372	-7.1	-1896.5	38.7
Node 16B	-1781	-153	-1292	-1.1	-276.7	9.0
Node 17B	-1913	878	-2736	-68.1	-1513.0	-112.1
Node 01C	-4097	-875	-4476	2447.4	-798.1	-131.9
Node 02C	-2222	-994	-4168	17.2	250.2	-68.5
Node 03C	-1332	919	-5155	26.3	72.0	-17.2
Node 04C	-1077	511	-6428	16.7	1185.0	-679.4
Node 05C	-717	330	-7710	9.3	-342.2	-43.9
Node 06C	-473	186	-5984	1750.4	-1218.2	69.6
Node 07C	1128	261	-4467	19.4	370.8	-15.5
Node 08C	-762	822	-4702	20.0	226.8	-17.4
Node 09C	-822	869	-4877	19.5	226.4	-21.4
Node 10C	-2577	-21	-10	0.6	188.3	-16.7
Node 11C	-2562	64	-112	0.8	274.5	-19.2
Node 12C	-2466	-67	103	-48.2	-57.4	-24.9
Node 13C	-1050	211	-3541	19.5	287.1	66.2
Node 14C	-869	-692	-3152	433.8	-388.5	5.4
Node 15C	-1058	1068	-2694	30.0	-150.4	-37.0
Node 16C	-2706	17	-4	-0.5	109.2	-19.1
Node 17C	-1626	421	-2790	20.8	146.9	14.7
Node 18C	-1829	-1429	-402	314.4	-1663.2	36.2
Node 01D	-1834	-478	606	105.5	-1219.4	-108.4
Node 02D	-339	-223	836	42.1	-475.5	29.4
Node 03D	-321	-115	-273	17.7	-514.9	22.9
Node 04D	-1954	-205	555	-71.3	-833.9	-81.9

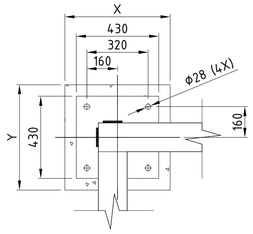
1.0 Roof Live Load

NODE	Fx (lb)	Fy (lb)	Fz (lb)	Mx (lb.m)	My (lb.m)	Mz (lb.m)
Node 01A	727	848	4067	1555.8	-1248.7	-5.0
Node 02A	1734	25271	51099	17268.1	532.2	1205.6
Node 03A	-442	723	7895	-1.7	-147.0	162.9
Node 04A	391	729	8799	-12.4	-510.8	71.5
Node 05A	-785	860	9607	-17.9	747.4	30.2
Node 06A	-436	2187	7499	1540.9	327.2	-74.2
Node 07A	194	860	6932	5.3	-256.5	-3.6
Node 08A	-218	398	7272	-6.4	-61.1	40.5
Node 09A	257	36	7581

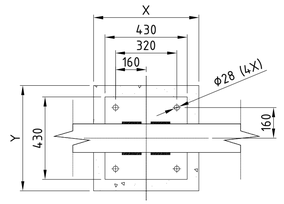
DRAWING NUMBER: MD-216-49PQ-EG-EL-LAD-V552-1001 | REV: A03



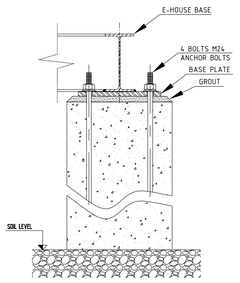
FRONT VIEW



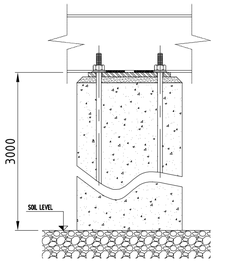
DETAIL A



DETAIL B



SECTION A-A



SECTION B-B

1.0 Dead Load

Node	Fx (N)	Fy (N)	Fz (N)	Mx (N.m)	My (N.m)	Mz (N.m)
Node 01A	981	608	34260	3265.1	-10720.5	102.3
Node 02A	-2045	11472	42463	201.3	39.7	-378.7
Node 03A	-879	7201	37081	185.0	1472.0	-406.1
Node 04A	341	6004	37858	148.3	-548.8	-398.7
Node 05A	1060	3906	26943	89.1	3615.7	-216.1
Node 06A	305	1634	29274	5191.8	-2761.9	102.5
Node 07A	-987	13669	26344	185.1	1167.7	-494.7
Node 08A	64	10310	27551	216.7	277.8	-469.8
Node 09A	-1558	9684	26983	223.6	-410.1	-128.7
Node 10A	-724	11305	30255	256.3	1418.2	265.4
Node11A	-1464	2793	25428	7880.5	6121.5	142.1
Node 01B	3686	-513	23227	564.0	-7476.2	35.6
Node 02B	262	-1724	54218	30.8	-143.5	192.2
Node 03B	1778	2376	65611	73.4	-3582.8	6.6
Node 04B	-1606	2969	70620	67.1	473.2	-123.0
Node 05B	0	3185	56031	43.9	2544.0	-100.9
Node 06B	4405	-1522	34867	27.3	4414.2	476.2
Node 01C	-1524	-358	34943	-3420.7	-2805.4	122.5
Node 02C	347	-9448	40893	-118.3	2598.8	306.0
Node 03C	1010	-10048	39380	-213.5	835.6	374.3
Node 04C	-1686	-9610	30699	-226.1	40.6	-102.8
Node 05C	-964	-11699	30963	-261.3	1070.3	-467.6
Node 06C	-1413	-2745	20807	-7707.0	6175.8	-180.2
Node 01D	2357	-2566	12800	-3000.4	-3804.0	-61.4
Node 02D	206	-8777	24486	-88.3	-456.4	118.3
Node 03D	164	-8493	23056	-90.1	-164.9	205.8
Node 04D	-121	-7919	24582	-84.9	-845.0	3.4
Node 05D	1086	-6131	34109	-87.7	-1046.5	138.7
Node 06D	-1138	20	13948	-578.6	5773.1	82.8
Node 01E	-1757	-1167	2391	-618.1	-1265.4	124.5
Node 02E	66	-2535	5644	-2357.5	1190.5	-174.6
Node 01F	-12	-1054	4070	-617.8	-1422.3	-54.3
Node 02F	-378	-1162	7188	-1235.5	1457.1	8.7

1.0 Live Load

Node	Fx (N)	Fy (N)	Fz (N)	Mx (N.m)	My (N.m)	Mz (N.m)
Node 01A	824	-37	17707	3087.3	-1580.6	155.1
Node 02A	-2219	10783	32567	202.2	620.8	-371.5
Node 03A	127	8402	31058	179.2	-386.3	-312.7
Node 04A	1044	7500	31450	167.1	-612.4	-230.9
Node 05A	3552	8160	32235	175.5	337.5	-113.8
Node 06A	-3890	-1943	44279	13168.0	-5402.6	155.4
Node 07A	-4681	54077	57721	876.1	-2543.0	-597.8
Node 08A	105	43883	73377	1037.0	1292.8	-1713.5
Node 09A	-5866	37779	75587	1024.3	1362.7	-469.4
Node 10A	-5980	39154	73826	962.3	3200.5	725.4
Node11A	-2164	3585	37975	15570.0	13860.0	410.3
Node 01B	7509	1413	55062	5351.9	-21890.0	-87.6
Node 02B	-615	13109	122180	122.8	531.2	-448.0
Node 03B	244	10806	118780	113.7	-1228.1	-332.0
Node 04B	-502	9560	119430	101.5	-1823.5	-243.6
Node 05B	2597	7094	108610	67.8	1758.3	-192.3
Node 06B	29831	-4328	93928	-6646.5	3109.7	1871.2
Node 01C	-9192	4305	36499	-5303.0	-1747.8	397.0
Node 02C	3053	-35149	123930	-418.3	-2689.1	1081.9
Node 03C	1236	-44087	81350	-1018.6	7171.8	1148.6
Node 04C	-5191	-37897	75487	-1011.4	1322.4	-306.1
Node 05C	-6337	-39994	73021	-970.3	3310.3	-1483.8
Node 06C	-1810	-3249	38000	-15550.0	13879.0	-545.1
Node 01D	2587	-2548	21107	-6837.1	-8791.0	-62.7
Node 02D	1468	-24174	46092	-382.0	576.8	675.1
Node 03D	2240	-19330	45465	-342.7	125.0	516.8
Node 04D	42	-17249	45527	-320.9	-428.0	117.1
Node 05D	6091	-13234	69503	-267.2	-8936.2	203.6
Node 06D	-3490	739	23998	-1269.3	13275.0	180.9
Node 01E	-10189	-2423	-9690	32153.8	-1774.4	582.9
Node 02E	885	-12583	-2755	-3001.6	1516.2	-442.8
Node 01F	-792	-1358	1750	864.3	-2553.1	-179.4
Node 02F	-17	-1206	10130	-3653.9	2693.0	14.0

1.0 Wind Load Transversal

Node	Fx (N)	Fy (N)	Fz (N)	Mx (N.m)	My (N.m)	Mz (N.m)
Node 01A	-953	-8102	-10260	3964.0	2359.6	-325.8
Node 02A	-2242	-2200	-11087	130.3	-265.3	-380.3
Node 03A	-3402	-2688	-12292	180.1	-140.2	411.0
Node 04A	-287	3413	-11746	387.7	-1028.1	418.2
Node 05A	3974	670	-12401	193.7	-211.2	-967.8
Node 06A	110	-12272	-16172	6456.2	-844.1	-2.4
Node 07A	-3878	-2862	-12622	167.4	576.0	-213.6
Node 08A	-6311	-7933	-13199	229.4	436.2	1388.5
Node 09A	-2717	-2306	-13142	670.4	153.2	1055.5
Node 10A	3211	-4480	-13022	274.2	22.2	-431.5
Node11A	3575	-18277	-14704	3183.7	-3438.8	500.9
Node 01B	1737	-13180	-5319	3876.7	2996.4	-137.0
Node 02B	3707	-14188	-8119	17.3	-4133.9	576.7
Node 03B	-755	-14249	-8578	74.3	4111.8	571.9
Node 04B	3867	-14808	-5136	109.7	-3676.4	252.3
Node 05B	13	-10510	-4176	69.2	2597.1	-86.2
Node 06B	2423	-14220	-3488	876.1	-739.4	199.9
Node 01C	2225	-8436	184	4403.7	7.3	-208.3
Node 02C	1285	-4799	-10730	40.6	-276.2	255.4
Node 03C	5465	-8827	-610	170.8	-2236.2	1087.0
Node 04C	-397	-4542	1611	376.4	1.2	261.1
Node 05C	-5913	-6902	1585	396.2	-450.0	-556.0
Node 06C	-5693	-17567	8303	8563.0	249.1	486.3
Node 01D	2823	-8413	7357	4652.2	-75.3	-348.3
Node 02D	2447	-1374	1398	134.4	51.8	-320.5
Node 03D	2575	-619	908	203.9	55.1	579.8
Node 04D	-186	3114	860	361.1	-11.8	-77.3
Node 05D	-5490	3085	-11093	174.7	3556.3	-560.4
Node 06D	-819	-4869	-2927	4439.1	-2546.8	38.2
Node 01E	2331	-9941	7280	4070.3	-800.6	-214.1
Node 02E	-1364	-13200	5843	5426.9	482.8	220.3
Node 01F	1623	-11925	8319	2975.1	-440.9	-90.4
Node 02F	-3023	-7143	6096	3365.3	164.2	163.5

1.0 Wind Load Longitudinal

Node	Fx (N)	Fy (N)	Fz (N)	Mx (N.m)	My (N.m)	Mz (N.m)
Node 01A	-3462	1171	-3750	-1160.1	-52.8	46.4
Node 02A	-1370	-358	-4620	-19.7	53.0	49.5
Node 03A	-857	-783	-5006	-26.7	222.8	-28.8
Node 04A	-898	-628	-4741	-33.8	-345.3	31.7
Node 05A	-827	-755	-4493	-28.4	-99.8	34.6
Node 06A	-928	1952	-3824	-1487.8	-969.0	24.6
Node 07A	-648	-857	-2988	-33.6	235.9	39.0
Node 08A	-1070	-1119	-3675	-34.5	-44.6	-28.7
Node 09A	-1021	-1692	-3754	-48.7	-74.9	-33.7
Node 10A	-963	-1722	-3954	-46.4	-44.4	15.5
Node11A	-1991	1721	-1102	-1252.8	-1589.4	16.5
Node 01B	-5359	-809	-6165	-325.5	-428.0	-80.8
Node 02B	-587	-609	-6449	-4.7	-3850.4	29.9
Node 03B	-3194	-280	-8855	-2.7	4329.8	-17.9
Node 04B	2242	-719	-8662	-5.3	-4235.9	47.8
Node 05B	-1420	21	-11067	-3.2	4104.5	14.7
Node 06B	111	1138	-6819	393.7	-1751.8	-49.8
Node 01C	-508	-1159	-3709	987.9	-302.4	-31.3
Node 02C	-1145	1189	-4423	11.3	287.9	-25.7
Node 03C	-800	1601	-4260	22.0	394.1	14.7
Node 04C	-1109	1701	-3730	53.6	-107.6	-8.7
Node 05C	-1223	1548	-3963	41.7	-22.0	8.4
Node 06C	-2455	-1798	-848	964.5	-1938.3	2.3
Node 01D	-3002	-1286	-3991	2013.4	-362.4	82.4
Node 02D	-1737	637	-5172	25.9	247.3	-43.7
Node 03D	-1040	841	-5769	28.3	216.1	21.6
Node 04D	-1019	1487	-5686	49.4	-233.8	-40.4
Node 05D	-1288	1115	-6997	34.9	61.4	-58.6
Node 06D	-175	-1070	-2608	714.3	-2199.6	-18.4
Node 01E	-143	-63	-952	-56.0	-311.5	-5.4
Node 02E	-1117	460	-202	445.8	-928.2	32.6
Node 01F	-1731	-704	-1420	-4.3	-725.1	23.6
Node 02F	-1291	-269	-928	256.2	-1257.8	22.5

1.0 Seismic Load X

Node	Fx (N)	Fy (N)	Fz (N)	Mx (N.m)	My (N.m)	Mz (N.m)
Node 01A	-1592	39	-2089	-303.1	-871.3	45.1
Node 02A	-2275	-13	171	0.0	183.3	7.6
Node 03A	-2048	22	7	-0.5	17.8	13.5
Node 04A	-2225	33	41	-0.8	-198.8	-3.6
Node 05A	-2179	-40	-132	0.0	58.6	9.7
Node 06A	-2608	-142	173	122.6	-1178.0	41.0
Node 07A	-2252	-48	3	0.5	43.9	11.8
Node 08A	-2189	10	13	0.2	39.8	19.6
Node 09A	-2334	45	2	0.2	-156.4	-0.4
Node 10A	-2427	-341	-595	-5.3	287.7	-8.4
Node11A	-2094	390	2418	513.2	-1639.4	76.8
Node 01B	-1239	-38	-1339	-6.8	-1684.3	-64.2
Node 02B	-1210	100	1456	0.2	515.0	-1.3
Node 03B	-627	12	-410	-0.2	123.7	-1.5
Node 04B	-617	33	375	0.0	173.6	-1.6
Node 05B	-788	-1	-859	-0.8	73.5	2.7
Node 06B	-1265	-35	677	196.3	-1288.8	-19.2
Node 01C	-1391	85	-1487	107.6	-1105.1	-27.2
Node 02C	-1776	-16	331	0.3	808.1	8.2
Node 03C	-1880	-2	-442	1.8	-302.1	-15.6
Node 04C	-2470	-60	23	0.4	-211.8	4.2
Node 05C	-2595	339	-710	5.6	379.5	15.4
Node 06C	-2220	-271	2548	-746.0	-1978.6	-63.6
Node 01D	-1647	3	-2302	-61.9	-1023.3	-59.6
Node 02D	-2277	-87	331	-2.1	294.5	-4.8
Node 03D	-2009	-11	29	-0.1	9.0	-16.4
Node 04D	-2248	-9	-23	-0.2	-231.3	3.1
Node 05D	-1880	-103	749	-2.2	-781.4	-7.3
Node 06D	-1157	80	1139	39.5	-1506.8	-34.5
Node 01E	-829	-82	-1316	-44.9	-731.2	-8.3
Node 02E	-822	35	1243	238.5	-711.4	-3.0
Node 01F	-847	63	-1284	-68.6	-648.0	-10.4
Node 02F	-827	11	1092	256.2	-668.0	-8.5

1.0 Seismic Load Y

Node	Fx (N)	Fy (N)	Fz (N)	Mx (N.m)	My (N.m)	Mz (N.m)
Node 01A	-420	-2231	-2691	1055.9	324.6	-85.6
Node 02A	-701	-94	-1597	35.7	-108.7	-129.4
Node 03A	-1062	-184	-1865	51.3	-46.2	116.7
Node 04A	-45	1451	-1774	101.3	-257.0	106.1
Node 05A	1043	619	-2078	53.5	-25.2	-272.7
Node 06A	140	-2686	-3724	2311.1	-265.8	0.9
Node 07A	-806	-178	-1513	40.1	64.9	-74.4
Node 08A	-1371	-1563	-1591	50.2	125.1	328.6
Node 09A	-397	-271	-1588	101.8	28.7	252.9
Node 10A	949	-751	-1520	57.4	53.7	-143.1
Node11A	911	-4153	-3168	1183.7	-424.1	115.2
Node 01B	290	-4672	-212	1478.7	38.9	-50.1
Node 02B	289	-3533	-72	4.8	53.6	138.6
Node 03B	389	-3823	448	19.1	-14.2	100.2
Node 04B	403	-3891	305	30.3	113.1	55.7
Node 05B	282	-3881	1306	19.0	-102.6	-32.9
Node 06B	263	-4489	194	340.1	211.2	62.0
Node 01C	631	-2691	826	1367.9	-86.0	-56.9
Node 02C	244	-1591	-2763	10.3	5.3	57.6
Node 03C	1433	-648	795	52.8	-793.6	324.9
Node 04C	15	0	1601	109.4	14.8	80.5
Node 05C	-1167	-1087	1527	54.5	-37.6	-157.6
Node 06C	-811	-4265	3186	2208.6	406.9	135.2
Node 01D	298	-2538	3283	1378.2	-412.1	-97.2
Node 02D	384	-202	1613	38.6	16.1	-115.1
Node 03D	698	378	1473	61.6	-81.9	183.6
Node 04D	-94	-2873	1440	113.8	25.8	-7.6
Node 05D	-1555	1423	-2838	55.1	1086.2	-213.0
Node 06D	-208	-1304	-499	1491.5	-747.0	18.5
Node 01E	693	-2856	2806	1424.2	-330.0	-58.2
Node 02E	-291	-3659	2500	1738.8	390.2	59.0
Node 01F	248	-3914	3272	1299.0	-389.6	-23.8
Node 02F	-733	-2472	2637	1380.0	246.1	66.4

1.0 Roof Live Load

NODO	Fx (N)	Fy (N)	Fz (N)	Mx (N.m)	My (N.m)	Mz (N.m)
Node 01A	-177	106	2395	394.8	-877.0	38.4
Node 02A	-206	359	5189	0.6	6.0	7.9
Node 03A	-97	386	5524	6.2	-101.9	8.3
Node 04A	-45	-487	5217	-11.9	254.7	-67.8
Node 05A	-100	223	4870	-0.3	158.3	84.3
Node 06A	74	918	4782	744.2	-65.2	10.6
Node 07A	138	840	6942	-10.6	-300.8	24.9
Node 08A	152	-97	7215	-19.9	9.0	-101.6
Node 09A	624	-1374	6923	-49.6	8.3	-55.0
Node10A	231	-579	6521	-28.2	380.1	80.0
Node11A	-840	1193	4213	1028.4	1314.6	12.1
Node 01B	1324	258	4604	250.1	-1632.6	13.1
Node 02B	-1801	357	7853	2.9	4121.3	-21.3
Node 03B	2221	-118	8168	2.4	-4071.7	28.3
Node 04B	2385	420	6482	5.0	4070.5	-40.0
Node 05B	1166	245	8578	1.0	-3902.3	4.6
Node 06B	-1456	-233	6950	-49.7	1348.2	8.4
Node 01C	381	-351	4269	-326.7	-712.0	-4.4
Node 02C	198	-346	6804	-8.8	483.7	-22.8
Node 03C	781	386	7167	23.5	-54.1	167.8
Node 04C	482	1508	6997	62.2	15.7	19.2
Node 05C	150	394	6573	27.9	399.2	-63.3
Node 06C	-841	-1372	4423	-1864.1	1182.6	-15.5
Node 01D	-438	-274	2922	-720.4	-1064.0	-13.4
Node 02D	-190	-838	6017	1.8	-57.2	-32.3
Node 03D	-14	-221	6232	8.4	-229.7	26.6
Node 04D	-120	125	6598	22.5	100.3	17.5
Node 05D	91	-271	6818	1.5	-117.0	-57.3
Node 06D	-298	-219	3379	85.6	1348.8	20.1
Node 01E	16	66	1419	-69.5	-267.1	38.3
Node 02E	-329	-392	3059	-1274.5	324.1	-55.9
Node 01F	141	-337	1135	-156.2	-329.1	-11.9
Node 02F	-231	-274	1572	-232.5	371.1	-6.8

GENERAL NOTES

1. ALL DIMENSIONS ARE IN MILLIMETERS UNLESS NOTED OTHERWISE
2. LOAD PARAMETERS FOR WIND AND SEISMIC CONDITIONS TAKEN FROM DOCUMENT MD-216-0000-EG-ST-CRT-0001
3. NUMBER OF PILLARS: 33
4. DIMENSIONS "X" AND "Y" SHOWED IN THE DETAILS "A" AND "B" SHALL BE DEFINED ACCORDING TO CONCRETE AND BUILD CHARACTERISTICS
5. CONCRETE FOUNDATIONS ARE NOT PART OF SIEMENS SCOPE
6. THE BOLTS SHOWED IN THE SECTIONS "A-A" AND "B-B" SHALL BE CAST INTO THE CONCRETE
7. ANCHOR BOLTS SHALL BE FABRICATED AS SPECIFIED IN DOCUMENT MD-101-SMEM-EG-CV-SPC-002
8. THE E-HOUSE BASE SHALL BE WELDED TO THE BASE PLATE ON SITE
9. THE CONCRETE COLUMNS SHALL BE FINISHED WITH 30 mm GROUT UNDER STEEL COLUMN BASE PLATE

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
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		A01	22/05/2020	ISSUED FOR INFORMATION	--	--	--	--	--

SIEMENS
Ingenuity for life

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DRAWING NUMBER	A1-MDN-C1907-E-1283-001	REVISION	0
DATE	14/05/2020	DATE	15/05/2020	DATE	16/05/2020
DRAWN	EDC	CHECKED	AC	APPROVED	VS

LARSEN & TOUBRO DRAWING NUMBER	019045_950_E_V_XXXXX_001		REVISION 2
OUTOTEC DRAWING NUMBER	900707__GAU01__WD01_EQS03__XXXXX	DOCUMENT ID	0U1006XXXXX	REVISION 2

DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	--	21/05/2020	LAYOUT
ENGINEER	--	--	CIVIL DESIGN INPUT
APPROVED	--	22/05/2020	FOUNDATION LAYOUT PREFAB SUBSTATION #04
EMPLOYER			
SCALE	1:XX	A1	DRAWING NUMBER: MD-216-49PQ-EG-XX-XXX-V552-XXX	REV: A03